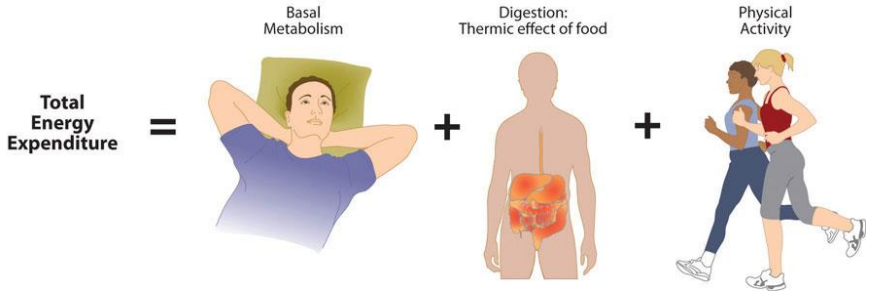
Nutritional status disorders

- Basal metabolic rate
- Nutrition
  - Carbohydrates
  - Proteins
  - Lipids
  - Vitamins
- Nutritional status assessment
  - Malnutrition
    - Marasmus
    - Kwashiorkor (kwashirkor-like syndrome)
  - Malabsorption

Basal metabolic rate

The lowest amount of energy expended to maintain vital processes in a waking state

- physical and mental calm
- fasting state (12-14 hours)
- thermoneutral environment



DAILY ENERGY OUTPUT
7 000-10 000KJ/day

JOJO efect

sleep -10%
long-term starvation - 40%

Growth
effort – up to 10x
regeneration

10% HIGHER RESTING METABOLISM

https://www.calculator.net/bmr-calculator.html

REGULATION OF METABOLIC PROCESSES

- **Nervous system** – autonomic nervous system.For example, **stimulation of the sympathetic nervous system leads to glycogenolysis**.
- **Liver**, **blood sugar regulation** (glucose storage, glycogen formation and breakdown, gluconeogenesis), **protein synthesis** (albumin, coagulation factors, plasma transport proteins, acute phase proteins, etc.), urea, lipids are stored in it, lipoproteins are formed there, cholesterol is excreted or synthesized into bile acids, it is a storage site for many vitamins, etc. Their detoxification function through hydroxylation reactions mediated by cytochromes P450 is significant, but it can be a source of toxic metabolites.
- **Digestive system** – intake of water and food, its digestion and utilization, and excretion of certain substances. Lipoproteins called chylomicrons are formed in the intestine. The intestinal microbiota also has a significant influence on metabolic processes.
- **Muscle mass** is an important site of energy consumption (glucose, fatty acids, ketone bodies) and a reservoir of proteins in case they are needed for the synthesis of other proteins (e.g., during inflammatory reactions or starvation). The main amino acid released by muscles is glutamine.
- **Adipose tissue** is an important energy store in the form of fat droplets. In adipose tissue, androgens are converted to estrogens, and many cytokines – adipokines – are produced that interfere with food intake (e.g., leptin) or metabolism (adiponectin, resistin).
- **Bones** (storage of substances, calcium reserves, buffering capacity for hydrogen ions).
- **Skin** – thermoregulation, storage of subcutaneous fat, metabolism of certain substances (e.g., production of vitamin D).
- **The kidneys** help maintain water and mineral balance and activate vitamin D through hydroxylation. Their excretory ("cleansing") ability is important, as is their ability to reabsorb important substances that are filtered into primary urine (glucose, amino acids).
- **The respiratory and cardiovascular systems** ensure an adequate supply of blood with oxygen and nutrients necessary for metabolism, as well as the removal of metabolic waste products.

Hormones

- **Insulin** increases glucose uptake into muscle and adipose tissue cells by affecting the membrane transporters for glucose GLUT4 and GLUT2 in hepatocytes. This reduces blood glucose levels, stimulates protein synthesis, glycogenogenesis, and lipogenesis, and inhibits lipolysis. Insulin suppresses appetite and induces the production of its antagonists in terms of their effect on blood glucose levels.
- **Glucagon** increases blood glucose levels by increasing glycogenolysis and gluconeogenesis. Its concentration rises physiologically during fasting and is pathologically elevated in glucagonoma tumors.
- **Growth hormone** (STH; somatotropic hormone) has a proteoanabolic effect. It increases glycemia by stimulating glycogenolysis and gluconeogenesis in the liver (causingthe development of insulin resistance) and also has a lipolytic effect.
- **Glucocorticoids** mobilize energy resources for vital organs through catabolismof proteins in muscles (and also in the skin) and lipolysis in subcutaneous fat. In the liver, they have an anabolic effecton glycogen synthesis. They increase blood sugar levels primarily by reducing glucose utilization(causing insulin resistance) and stimulating glucogenesis in the liver. They are stress hormonesand are partly responsible for metabolic changes that occur in response to stress, including insulin resistance.

Hormones

https://www.youtube.com/shorts/IgB-6l4_UJg

- **Thyroxine (T4) and triiodothyronine (T3),** which is several times more effective, cause an overall increase in metabolism,increase oxygen consumption and heat production in most organs, sensitize tissuesto the effects of catecholamines (thus promoting lipolysis), and reduce blood cholesterol concentrations. Physiological concentrations have rather proteoanabolic effects, but excessive amounts of thesehormones have a catabolic effect.
- **Sex hormones** have a predominantly anabolic effect, especially androgens (increased muscle mass,positive nitrogen balance).
- Estrogens inhibit bone resorption, affect the subcutaneous distribution ofadipose tissue, lead to sodium and water retention, and interfere with cholesterol metabolism (protectiveeffect against atherosclerosis).
- Progesterone has a thermogenic effect (increase in body temperatureduring the luteal phase of the ovarian cycle).
- **Cytokines, tissue hormones, and neurotransmitters** interfere with metabolism at many levels.These include appetite regulators (e.g., leptin, neuropeptide Y), gastrointestinalhormones (cholecystokinin, etc.), adipokines (cytokines produced in adipose tissue), and inflammatorycytokines (e.g., IL-1, IL-6, TNF).

1. carbohydrates



- 1g – 17kJ
- In the diet 60% (50-80%)
- 3-5g/kg/day

simple carbohydrates 1/6

- monosaccharides – glucose, fructose, galactose
- disaccharides – sucrose, lactose, maltose
- fruit, honey, sugar cane and beet, milk, confectionery, sweetened beverages

Complex carbohydrates 5/6

- polysaccharides – glycogen, starch, fiber
- potatoes, grains, legumes, vegetables, whole grain products

Polysaccharides representing fiber (cellulose, hemicellulose, mucin, pectins, and others), which is not absorbed (and therefore is not a source of energy), but has a number of beneficial effects on intestinal activity (e.g., slows down the absorption of carbohydrates, reduces cholesterol absorption, and prevents constipation).

Functions of carbohydrates

- IMMEDIATE ENERGY SOURCE (MAINLY BRAIN AND ERYTHROCYTES) (glucose, fructose)
- STORAGE SUBSTANCES (glycogen)
- STRUCTURAL FUNCTIONS (glycoproteins, glycolipids)
- COMPONENTS OF SOME MORE COMPLEX SUBSTANCES (nucleic acids, hormones, coenzymes)

- de novo synthesis from glycerol and AMK

Indigestible complex sugars - fiber

Soluble

- **absorbs water, swells – feeling of satiety**
- ferments
- nutrient for microbial flora – prebiotic
- Formation of usable MK = source of energy
- regulates the digestion of fats and carbohydrates
- Pectin

1:3

Insoluble

- **does not ferment, is not a source of energy**
- increases volume in the intestines -> accelerates the passage of waste and toxic substances, slows down the absorption of cholesterol
- lignin, cellulose



20-35g daylly

Glycemic index

- the rate at which glucose is utilized by the body from a given food
- the area under the glycemic curve during two hours after ingestion of a given food, as PERCENTAGE OF PURE GLUCOSE INTAKE – GLUCOSE =100
- Influenced by:
  - fiber content, food preparation, amount of chyme in the stomach

better utilization of glucose

DM type 2, obesity

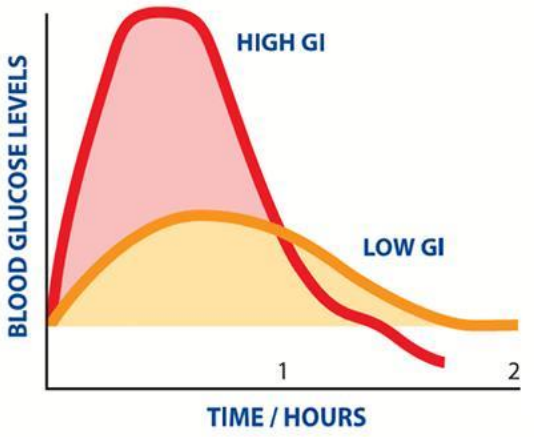
Low GI less than **55**

fruit, vegetables, nuts, whole wheat bread ...

Low GI less than 55

High GI – more than **70**

cornflakes, boiled potatoes, honey, chips, beer ...not suitable for diabetics



The amount of carbohydrate in the reference and test food must be the same.

2. proteins

- 1g = 17kJ (23kJ - urea)
- 10-15%, 0,8g/kg/day

AA → **ESSENTIAL**, **SEMI-ESSENTIAL**, **NON-ESSENTIAL**

Oligopeptides (2-10AA)

Polypeptides (11-50/100AA)

Vlastní bílkoviny (up to 50/100AA)

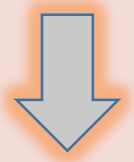


Function

- **VESSEL FILLING** albumin
- **BUILDING** collagen, elastin, keratin
- **TRANSPORT AND STORAGE** hemoglobin, transferrin
- **ENSURING MOVEMENT** actin, myosin
- **CATALYTIC, CONTROLLING, REGULATING** enzymes, hormones, receptors…
- **PROTECTIVE, DEFENSIVE** immunoglobulin, fibrin, fibrinogen

The quality of proteins depends on their amino acid composition. A complete protein diet is one that contains all essential amino acids (lysine, leucine, isoleucine, valine, methionine, phenylalanine, tryptophan, threonine, and conditionally essential histidine and arginine). High-quality proteins must be easily digested and amino acids absorbed by the body. Amino acids consumed in excess are not stored in the body, but are degraded by metabolic processes (formation of urea), the carbon skeleton is used for the synthesis of glycides (gluconeogenesis) and fats, and oxidized to produce energy.

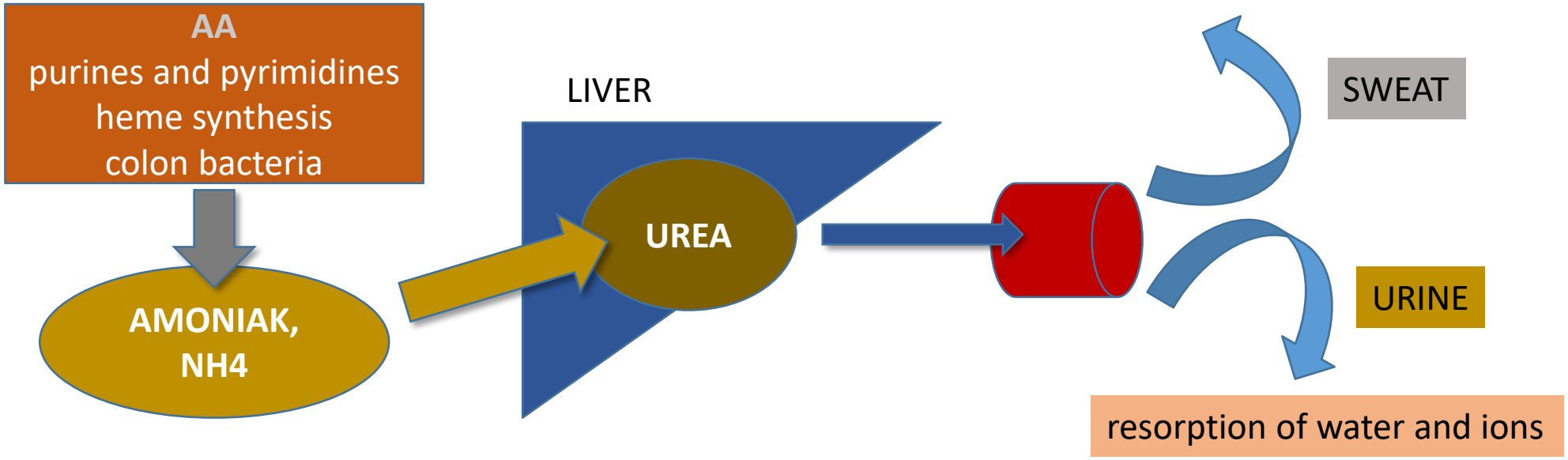
Proteins – **Dietary Reference Intake** (DRI) 15%

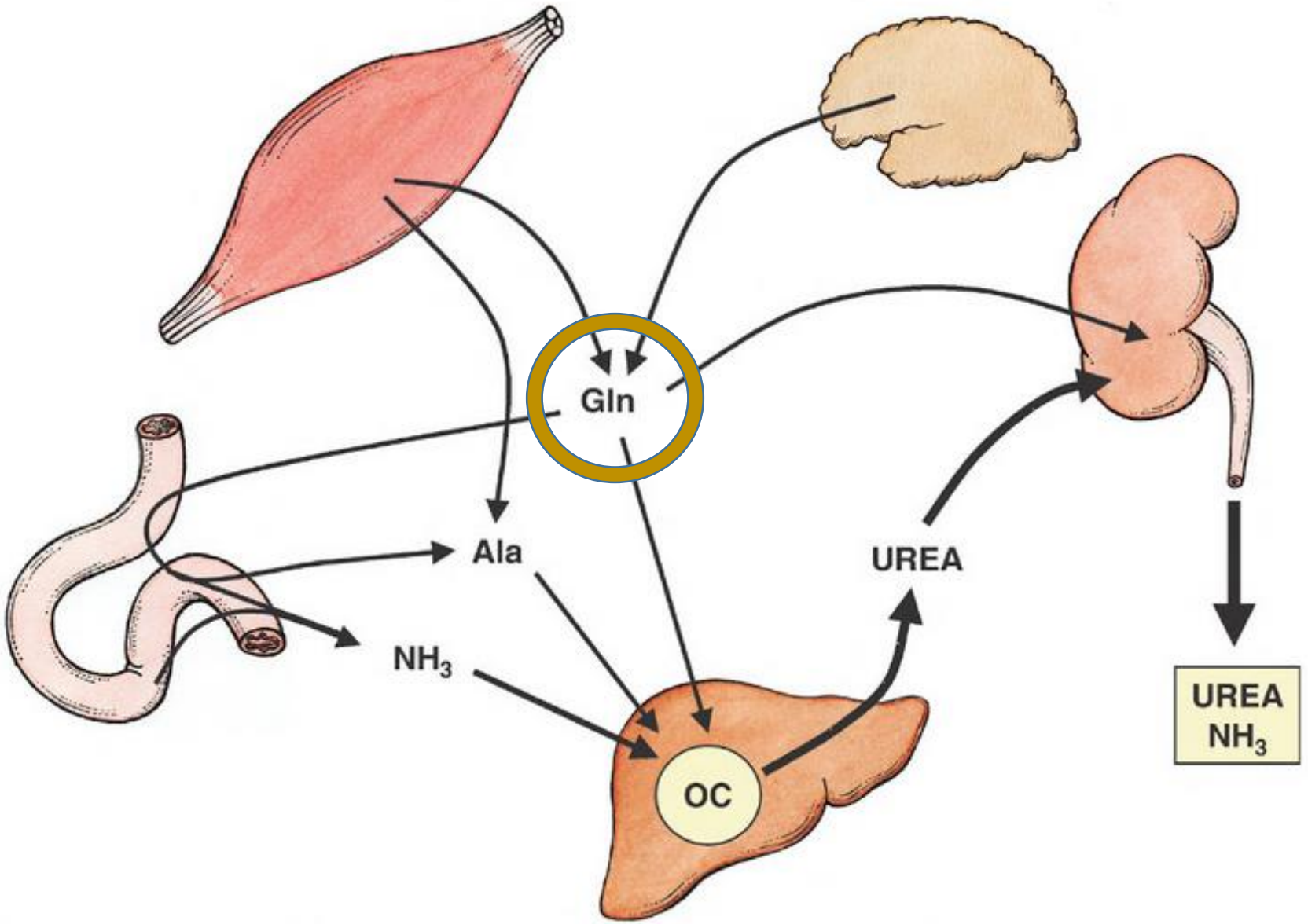
Childern	Adult	Seniors	Nursing mothers	Athletes
• **0,9-2,7** g/kg/day	• 0,8 g/kg/day 	• 1,0-1,2 g/kg/day	• 1,5 g/kg/day	• **1,6-2,2** g/kg/day (on anabolic steroids even more)

Liver and kidney disease - individually depending on health condition

Proteins - ureasynthetic cycle

- urea
- uremia (normal blood level 1.7-7.3mmol/l)
  - increased protein intake
  - catabolism
  - kidney failure





https://books.google.cz/books?id=XUkhYLAC3HwC&pg=PA170&lpg=PA170&dq=synt%C3%A9za+hemu+a+produkce+amoniaku&source=bl&ots=xlfusD_egN&sig=hYJrVOYr4_keIgAVcalQzLhy3VA&hl=cs&sa=X&ved=2ahUKEwi6n86E--7dAhWDh6YKHZmOAe0Q6AEwAnoECAcQAQ#v=onepage&q&f=false

3. lipids

- 1g = 38kJ
- 0,7–1,2g/kg/day
- < 30%

Potrava – FA from:
triacylglyceroles (90%)
phospholipides
Cholesterol esters



Simple — fats and oils, waxes

Composite — Phospholipids, glycolipids, lipoproteins

Lipids are the most energy-rich nutrient, a carrier of fat-soluble vitamins and a source of essential fatty acids.

All cells in the body, with the exception of cells of the central nervous system and erythrocytes, can directly use fatty acids as a source of energy.

The composition of lipids in food, especially the content of saturated or unsaturated fatty acids, plays an important role.

Function:

CELL MEMBRANES

SOLVENT OF VITAMINS

COMPONENT OF HORMONES

cholesterol

FORMATION OF GIT

THERMAL INSULATION

ENERGY RESERVE more than 50,000 kJ

4. Minerals

Calcium, Ca Formation of bones and teeth, participates in nervous activity

Magnesium, Mg Helps in bone formation, participates in nerve activity

Draslik, K Maintains ion balance, participates in nervous activity

Sodium, NaMaintains ion balance, participates in nervous aktivity

5. MICRONUTRIENTS AND TRACE ELEMENTS(**Fe, I, Cu**, Se, Cr, Mn, Mo, Co, F..)

Fe - An essential part of hemoglobin

I - Essential for thyroxine (thyroid hormone)

Cu – collagen, elastin

Se – antioxidant, THYROID hormone metabolism, deficiency – cardiomyopathy

Cr – stimulatory effect of insulin

Zn – forms the core of the insulin molecule

Mn – oxidative phosphorylation (metabolism of fats)

Mo – metabolism of purines

Co – B12, leafy vegetables

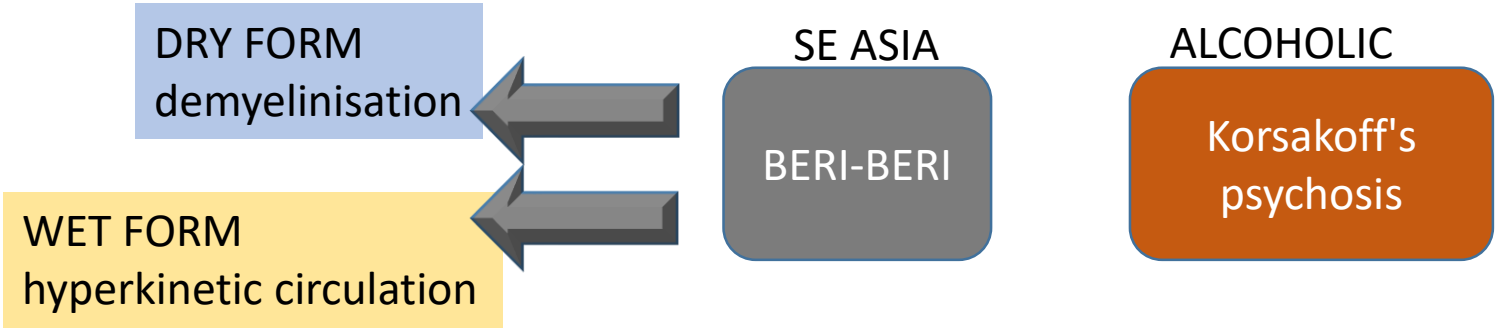
6. Vitamins

- The organism is dependent on the intake of vitamins or their precursors through food. Only small amounts of some vitamins can be formed endogenously: vitamin D from steroid precursors, vitamin K and biotin by intestinal microflora.
- For **fat-soluble (A, D, E, K)**, maldigestion or malabsorption of fats leads to combined hypovitaminosis; reserves in body fat (especially in the liver) prevent the development of deficiency symptoms for a long time (this applies especially to vitamins A and D); with excessive intake, their accumulation occurs and toxic manifestations of hypervitaminosis may occur.
- **Water-soluble vitamins (B, C)** are in minimal reserves in the organism and with excessive intake, they are rapidly excreted by the kidneys and do not accumulate in the organism, on the contrary, hypovitaminosis develops faster for the same reasons. The exception is vitamin B12, whose reserves in the liver are sufficient for several years.
- Hypovitaminosis
  - insufficient intake of the vitamin through food; e.g. vitamin D deficiency in breast milk, B1 deficiency in areas where the main food is peeled rice, vitamin B12 deficiency
  - insufficient endogenous vitamin production; e.g. intestinal dysmicrobia, lack of UV radiation
  - impaired resorption (impaired emulsification of bile fats, malabsorption syndrome of various etiologies, selective impaired resorption of vitamin B12 in pernicious anemia)
  - congenital or acquired disorder of vitamin metabolism at the level of its storage, transport or conversion into the active form (e.g. vitamin D-resistant rickets type I with a defect in the formation of 1,25-(OH)2-vitamin D3 in the kidneys)
  - resistance to the vitamin (e.g. vitamin D-resistant rickets type II in the absence of the receptor for 1,25-(OH)2-vitamin D3)
  - increased need for the vitamin (during growth, pregnancy, hyperthyroidism, malignancies, etc.)

Deficiency of water-soluble vitamins



- Vitamin B1 (thiamin)
- is lost during processing (peeled rice, white sugar)
- Vitamin B1 is active in the form of thiamine pyrophosphate and acts as a cofactor in the oxidative decarboxylation of γ-keto acids. maintains normal function of peripheral nerves.



• polyneuropathy (dry form) caused by degradation of the myelin sheaths of motor and sensory nerve fibers, especially in the lower extremities; it is manifested by paresthesias, muscle weakness and hyporeflexia

• cardiovascular syndrome (wet form) with hyperkinetic circulation caused by vasodilation of arterioles; edema may be present, the heart may (but may not) be extremely dilated

• Wernicke-Korsakoff syndrome is an encephalopathy with ophthalmoplegia, nystagmus, ataxia, apathy, disorientation and even psychotic disorders. There are small hemorrhages and degenerative changes of neurons in some parts of the brain.

- **Vitamin B2 (riboflavin)**
- Riboflavin, as a part of the coenzymes flavin mononucleotide and flavin adenine dinucleotide, participates in a number of oxidation-reduction reactions
- Vitamin B2 deficiency is associated with a deficiency of other vitamins and proteins-> developing countries
- In alcoholics and in patients with chronic infections or advanced malignancies.
- Deficiency is manifested by cheilosis (rhagades of the corners of the mouth, secondarily infected), atrophic glossitis, keratitis and dermatitis.
- source: meat, milk, eggs, yeast, vegetables

- **Vitamin B6 (pyridoxin)**
- This coenzyme is a cofactor for a number of transamination, carboxylation and deamination enzymes in amino acid metabolism and glycogen, necessary for the production of serotonin
- Vitamin B6 is present in practically all foods, but it can be lost during their processing.
- Secondary hypovitaminosis occurs with long-term use of drugs that act as pyridoxine antagonists (antituberculosis drug isoniazid, estrogens, penicillamine).
- Alcoholics suffer from vitamin B6 deficiency because acetaldehyde, potentiates the degradation of pyridoxine.
- The need for vitamin B6 is increased during pregnancy.
- The symptoms of vitamin B6 deficiency are similar to hypovitaminosis B2 – seborrheic dermatitis, cheilosis, glossitis, peripheral neuropathy.

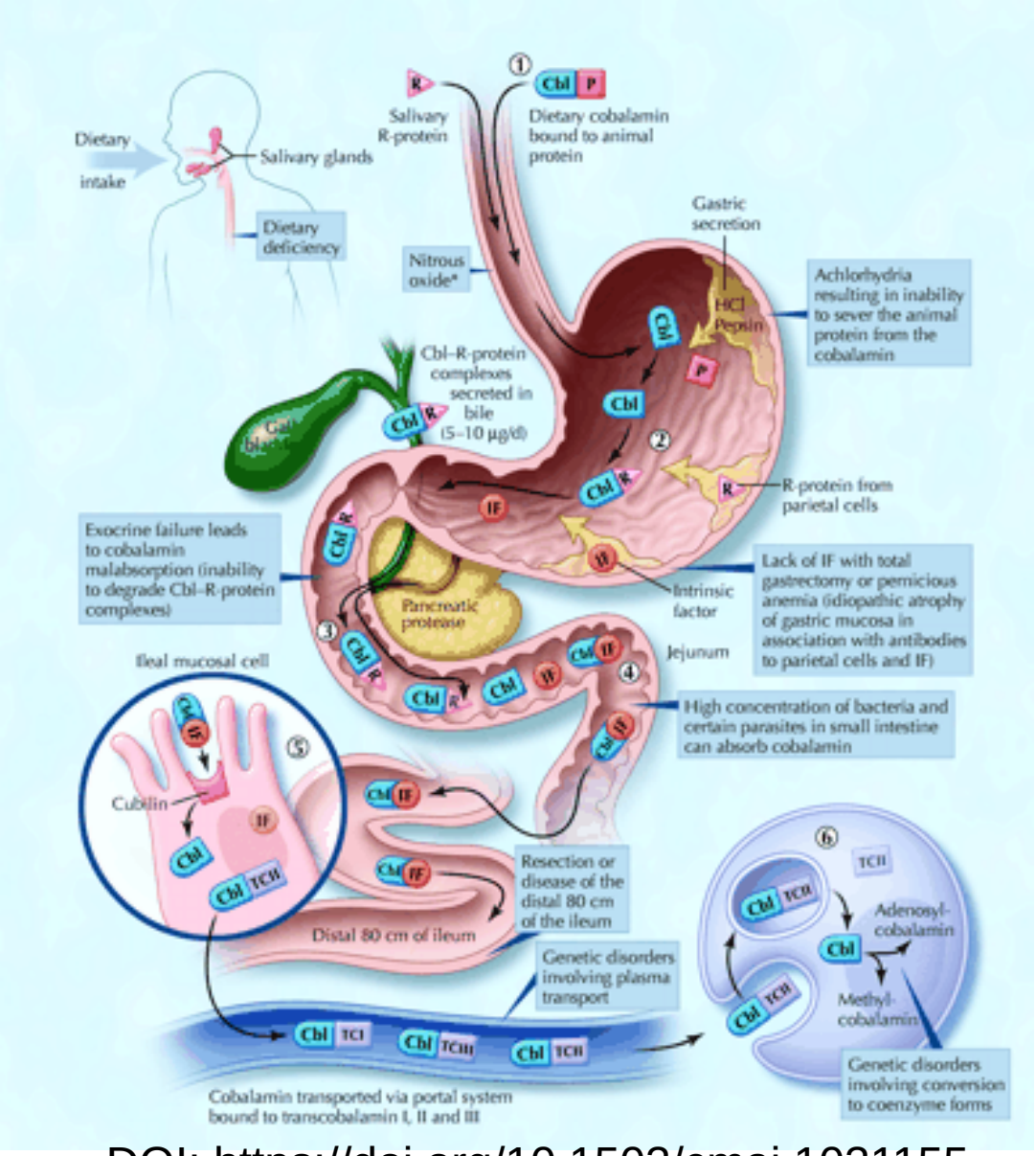
Vitamin B12 (cobalamin)

= complex of organic compounds containing a cobalt atom inside a circular molecule

Absorption:

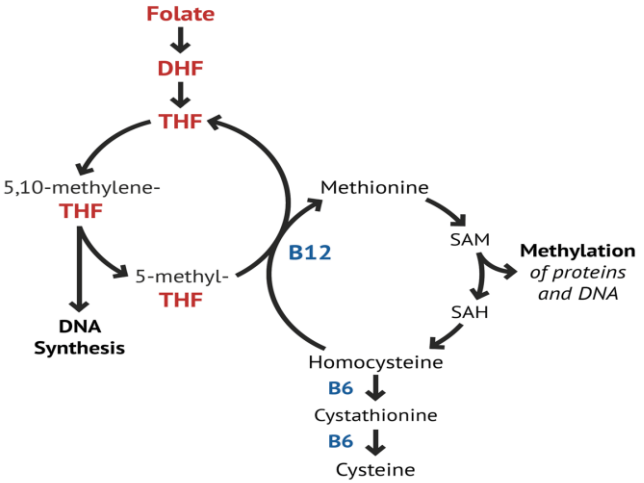
- In the stomach, binding to the GP group complex – R-binder
- in the duodenum, cleaved and bound to IF (intrinsic factor)
- in the distal ileum, resorbed after binding to specific receptors – cubilin -> by endocytosis

Total reserves of 4mg (1/2 in the liver, the rest in the bone marrow and elsewhere) last for 3-6 years



DOI: https://doi.org/10.1503/cmaj.1031155

Deficiency – causes + symptoms



- Vitamin B12 deficiency is primarily a consequence of malabsorption
- In our conditions, the main cause is a deficiency of intrinsic factor (IF) due to an autoimmune reaction against the parietal cells of the stomach -> atrophic gastritis with achlorhydria-> pernicious anemia
- IF deficiency occurs after gastrectomy and some bariatric operations and with reduced HCl production in the stomach
- anatomical intestinal lesions (strictures and fistulas)
- pseudo-obstructive intestinal conditions with significantly reduced intestinal motility (neuropathy in diabetes mellitus, scleroderma) -> massive colonization of the small intestine by bacteria that consume vitamin B12
- resection of the terminal ileum (Crohn's disease)
- MTHFR polymorphism-> enzyme with reduced activity. This may predispose to higher levels of homocysteine, which is one of the risk factors for atherosclerosis.

impaired DNA synthesis and cell division

- MEGALOBLASTIC ANEMIA
- leukopenia, thrombocytopenia
- glossitis, cheilosis, atrophy
- anorexia, diarrhea, dyspepsia
- Deficiency of methionine -> DEMYELINIZATION
  - paresthesias in the extremities, later weakness, ataxia, sphincter disorders, slow reflexes (peripheral neuropathy and spinal cord degeneration) to severe dementia or psychosis

neuroanemic syndrome

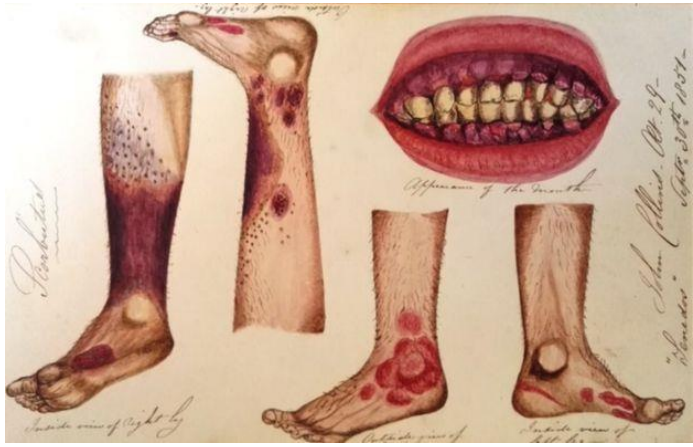
- These neurological disorders can also occur in patients without hematological symptoms

Folic acid B9

- mainly found in vegetables and fruit
- only small folate stores, so a constant and sufficient intake through food is necessary
- increased need – pregnancy (cleft lip, cleft palate, neural tube), growth, hemolytic anemia, cancer
- most common causes of hypovitaminosis include insufficient intake, especially in alcoholics
- Hypovitaminosis due to malabsorption is part of malabsorption syndromes (gluten enteropathy, tropical sprue).
- Folate from the diet must first be converted into its active form - tetrahydrofolate (THF). This reaction is catalyzed by the enzyme dihydrofolate reductase, which can be inhibited by some drugs (e.g., methotrexate), and its blockade can induce symptoms of folic acid hypovitaminosis
- megaloblastic anemia, glossitis, cheilosis, dyspeptic disorders, neurological symptoms are not present

Vit. C (Ascorbic acid)

- in milk, animal products (liver, fish), fruits and vegetables
- Antioxidant, hydroxylation of procollagen
- elderly people living alone or in alcoholics, peritoneal dialysis or hemodialysis
- *Def:* ↓collagen formation, ↓Fe absorption
- Wound healing is impaired and scar strength is reduced
- Vitamin C deficiency in its advanced form is referred to as scurvy and is manifested by bone disorders in growing individuals (osteoid matrix formation disorder, Möller-Barlow disease), subperiosteal hematomas, skin hemorrhages, gingivitis, and recurrent bacterial infections

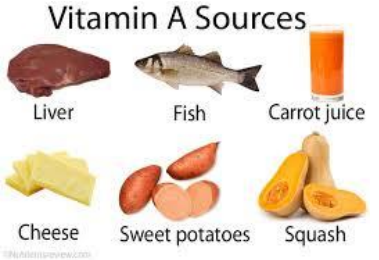


http://www.bbc.com/news/uk-england-37320399

Deficiency of fat-soluble vitamins

Vitamin A (retinol, retinal, retinoic acid and carotenoids – provitamins A)

- food of animal origin (liver, fish, eggs, milk), vegetables (spinach, carrots) contain carotenoids (carotenes)
- digestion and absorption of carotenes requires the presence of bile and pancreatic enzymes
- 90% of vitamin A in the liver (6 months)
- part of the pigments rhodopsin and iodopsin in the retina -> vision adaptation to lack of light
- EPITHELIAL DIFFERENTIATION – MAINLY DURING EMBRYOGENESIS
- ANTIOXIDANT



Lack:

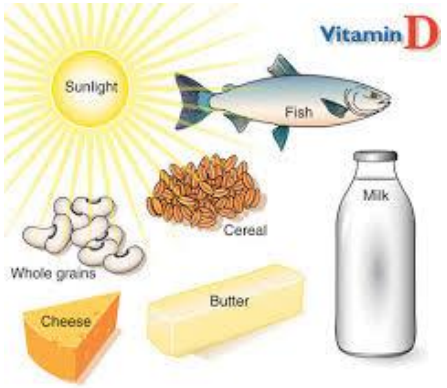
- night blindness
- Keratinization of the corneal and conjunctival epithelium is manifested by their dryness (xerophthalmia), the formation of erosions and even destruction of the cornea (keratomalacia) with loss of vision
- squamous metaplasia of the epithelium and its keratinization
- respiratory tract epithelium loses its protective antibacterial properties ->relapses of bronchitis and bronchopneumonia
- urinary tract epithelium ->higher frequency of urolithiasis

Excess:

- toxic manifestations
- Acute -headaches, vomiting, stupor
- Chronic -weight loss, vomiting, bone and joint pain, hyperostosis, and hepatomegaly
- pregnancy – teratogen in smokers: risk of tumor formation (bronchogenic cancer)

Vitamin D

- skin exposed to ultraviolet radiation (vitamin D3) 80%, sea fish and vegetables
- In the plasma bound to D-binding protein



-> liver->conversion to 25-hydroxyvitamin D

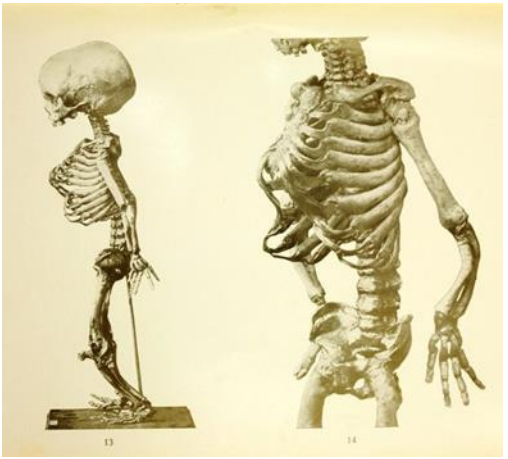
->kidneys-> active form 1,25-dihydroxyvitamin D (1,25(OH)2D(steroid agent)

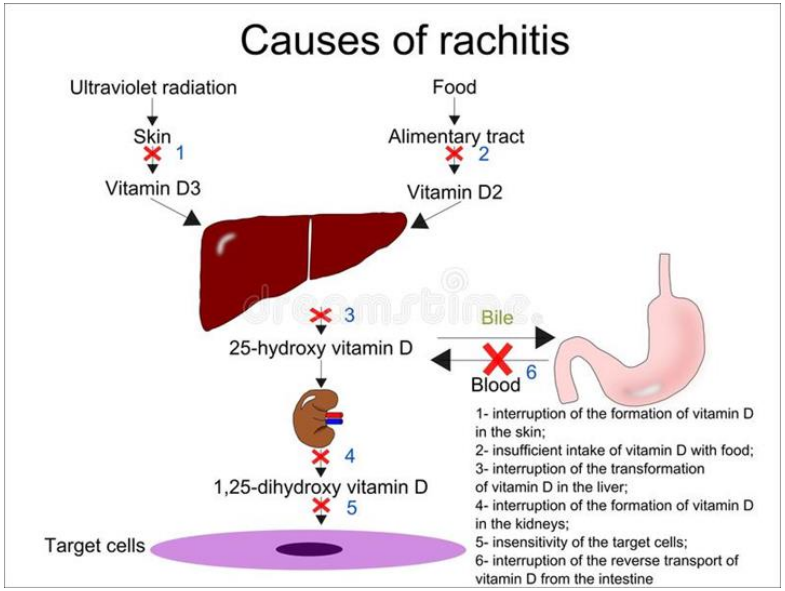
->binds to specific intracellular receptors (intestine, bones, skin and kidneys)-> maintain normal plasma calcium levels (neuromuscular activity, phosphate levels)

- 1,25(OH)2D stimulates the intestinal absorption of calcium and phosphates, together with parathyroid hormone mobilize bone calcium-> parathyroid hormone-dependent reabsorption of calcium in the distal tubules of the kidneys

Hypovitaminosis:

- combination of vitamin D deficiency in the diet and lack of UV radiation, peripheral resistance is rare
- impaired resorption in the intestine (cholestasis, pancreatic insufficiency, celiac disease)
- insufficient hydroxylation (hepatic, renal insufficiency, congenital vitamin D-dependent rickets type I)
- impaired ossification of newly formed osteoid and an excess of unmineralized matrix
- During grow-skeletal deformations (skull deformation, rachitic rosary of ribs, saber-shaped tibia, triangular shape of pelvic bones) RACHITIS: COMPLETE DISORDER OF SKELETAL MATURATION AND FORMATION
- Adult- insufficient mineralization interferes with the normal course of bone remodeling -> bone is weak, risk of fractures (osteomalacia).
- Laboratory hypocalcemia- increased bone turnover
- reduced resistance to infectious diseases -> metabolic syndrome and autoimmune diseases, including type 1 diabetes.





Hypervitaminosis

- hypercalcemia
- prolonged, causes excessive bone resorption
- Endogenous vitamin D production is increased in some granulomatous diseases (sarcoidosis)

https://www.dreamstime.com/stock-illustration-causes-rachitis-most-common-image69147405

Vitamin E



- a group of eight similar lipophilic compounds
- abundantly present in food (vegetables, meat)
- absorption is dependent on bile and pancreatic enzymes->resorbed vitamin E is transported into the blood in chylomicrons and becomes part of lipoproteins, especially LDL->is stored in a number of tissues, especially in fat, but also in muscles and the liver
- Hypovitaminosis due to malnutrition is not known in developed countries
- antioxidants -scavengers of free oxygen radicals.
- *prevention of atherogenesis ? (pro and con)*
- gamete formation

Hypovitaminosis:

- in lipid malabsorption (cholestasis, pancreatic insufficiency, primary diseases of the small intestine)
- in the rare abetalipoproteinemia
- nervous system disorders (reflex disorders, position and vibration sensation, pain perception, muscle weakness, vision disorders).
- decreased immunity, infertility
- in newborns anemia

Vitamin K

- synthesized by the intestinal flora, small exogenous intake through food is necessary
- cofactor of hepatic microsomal gamma-glutamyl carboxylase, which converts glutamyl residues to γ-carboxyglutamate
- creates a binding site for calcium
- VIT. K DEPENDENT PROTEINS

carboxylation
Ca2+ binding

- essential for coagulation (anticoagulants) factors (II, VII, IX, X, proteins C and S)
- Needed for carboxylation of osteocalcin present in osteoid
- In the absence of vitamin K, coagulation factors are still produced in the liver, but in a non-functional form

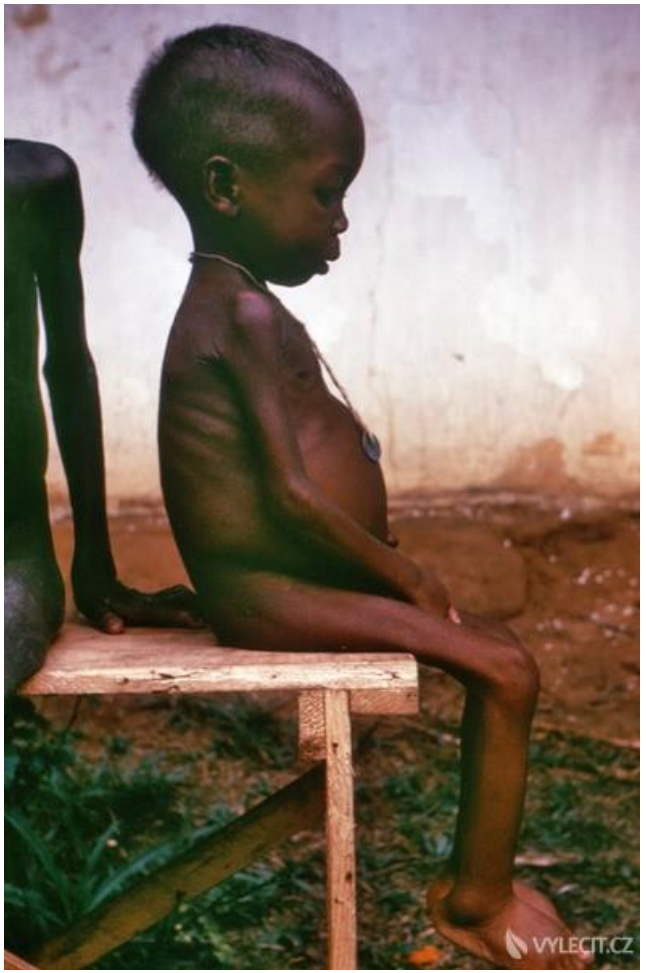


Lack:

- deficiency due impaired fat resorption, liver failure or drug-induced hypovitaminosis
- Iatrogenic induced by dicumarol preparations (warfarin)
- Long-term treatment with dicumarol -> higher risk of developing osteoporosis
- BLEEDING

NUTRITION

MALNUTRITION



OBESITY

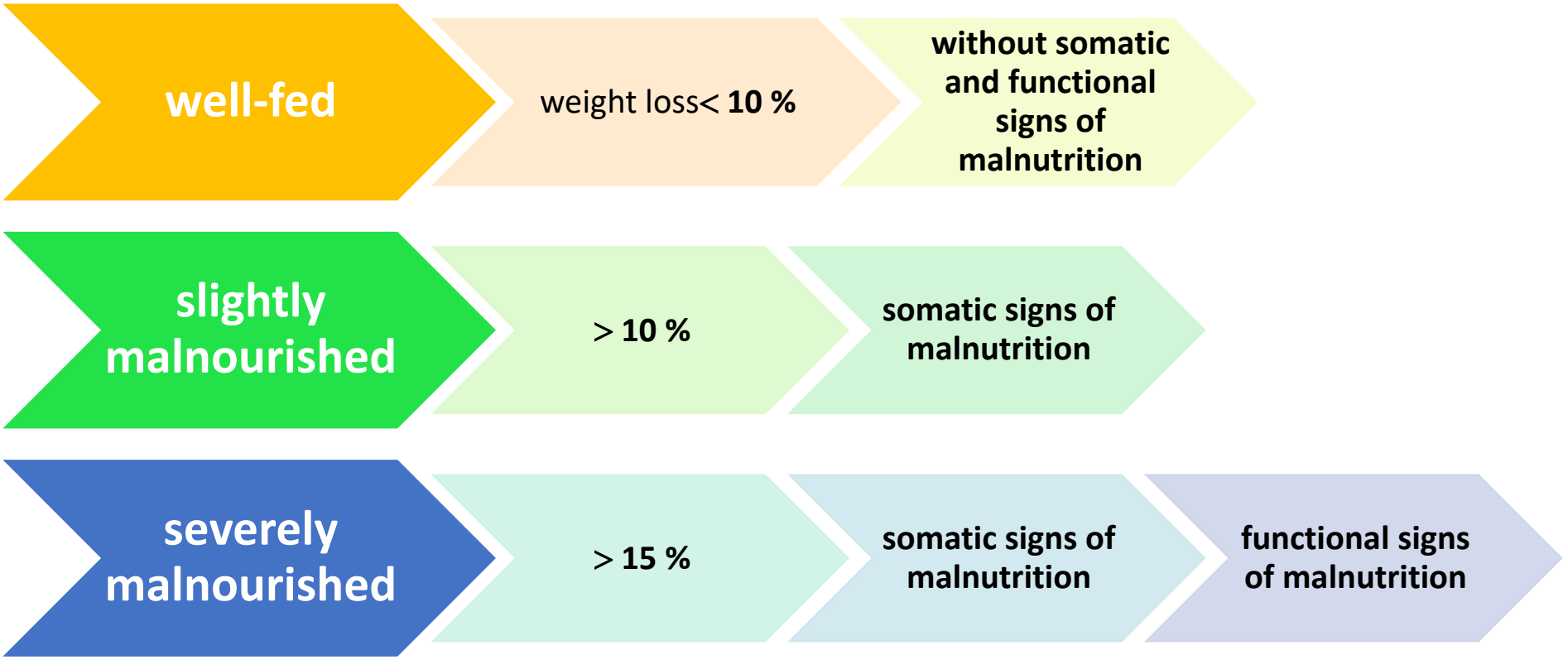


QUALITATIVELY UNBALANCED FOOD

Malnutrition diagnostics

SGA = subjective global assessment

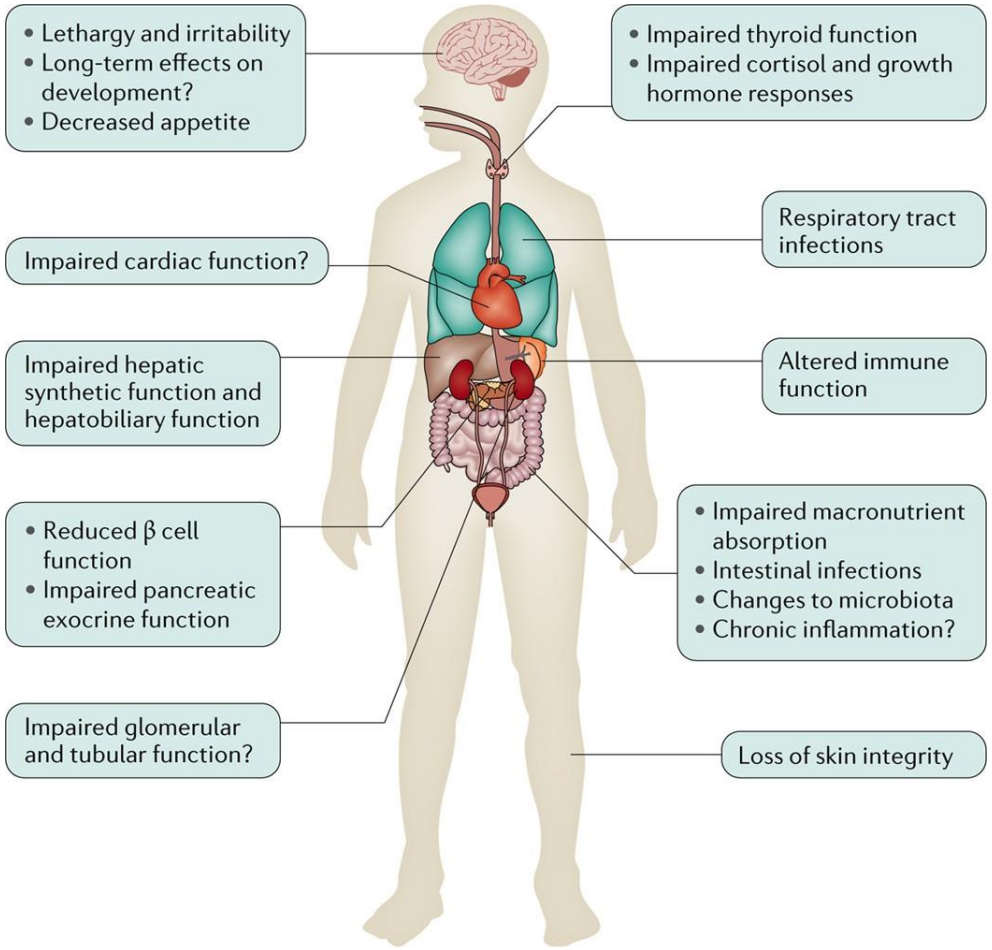
- patient eats a not healthy diet in which one or more nutrients are either not enough or are too much
- The aim is to identify malnourished patients or patients at risk of malnutrition/ 6 months



Somatic: Loss of muscle mass: temporomandibular region, limb muscles, Loss of body fat, Skin changes: loss of skin turgor, peeling skin, hair color changes, Swelling, hepatomegaly (steatosis)

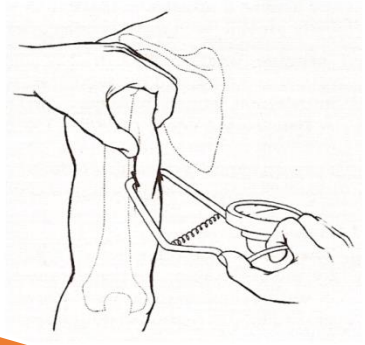
Functional: immunodeficiency, atrophy with reduced function – bradycardia, hypotension

Malnutrition



reduced appetite
Infertility
lack of interest in food and drink
hypothyroidsm
feeling tired all the time
feeling weaker
getting ill often and taking a long time to recover
anemia
wounds taking a long time to heal
poor concentration
feeling cold most of the time
low mood, sadness and depression

1. Assessment of fat stores



Anthropometry

BMI

waist circumference

mid-arm circumference

skinfold



Bioimpedance

low intensity, high frequency current



Ultrasound

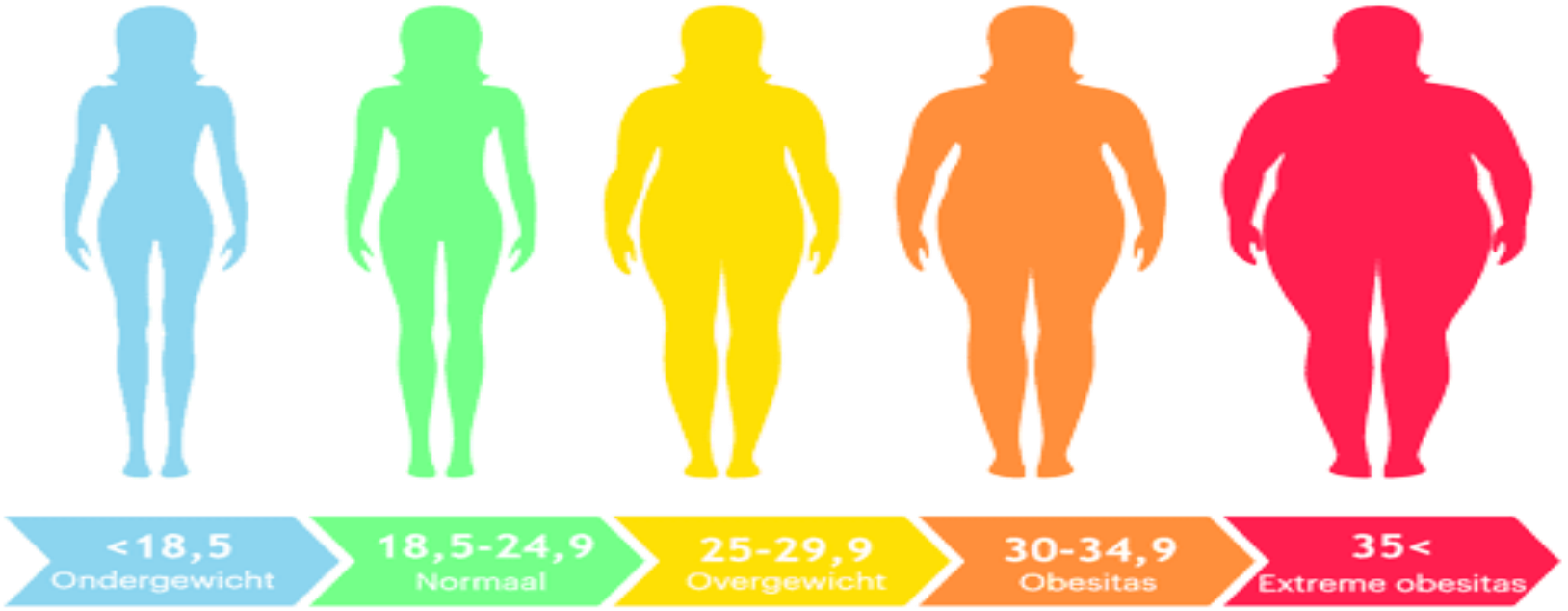
visceral fat

Body mass index (BMI)

Obesity is defined as a body mass index (BMI) that exceeds 30 kg/m2 and generally develops when caloric intake exceeds caloric expenditure in genetically susceptible individuals.

Waist-to-hip ratios is useful in visceral and peripheral obesity

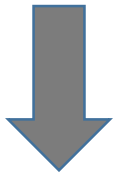
Obesity is a risk factor for hypertension, stroke, hyperlipidemia, gallstones, nonalcoholic steatohepatitis, gastroesophageal reflux, osteoarthritis, infectious disease, asthma, and obstructive sleep apnea.



2. Assessment of protein reserves

- Body protein
- muscle (30%)
- visceral (70%) prealbumin, albumin, transferrin, …

- I. MEASUREMENT OF ARM CIRCUMFERENCE
- II. CREATININE
- is formed from creatine only in muscle
- is not reused
- is excreted only in urine
- increased excretion during physical exertion, acute illness, meat intake

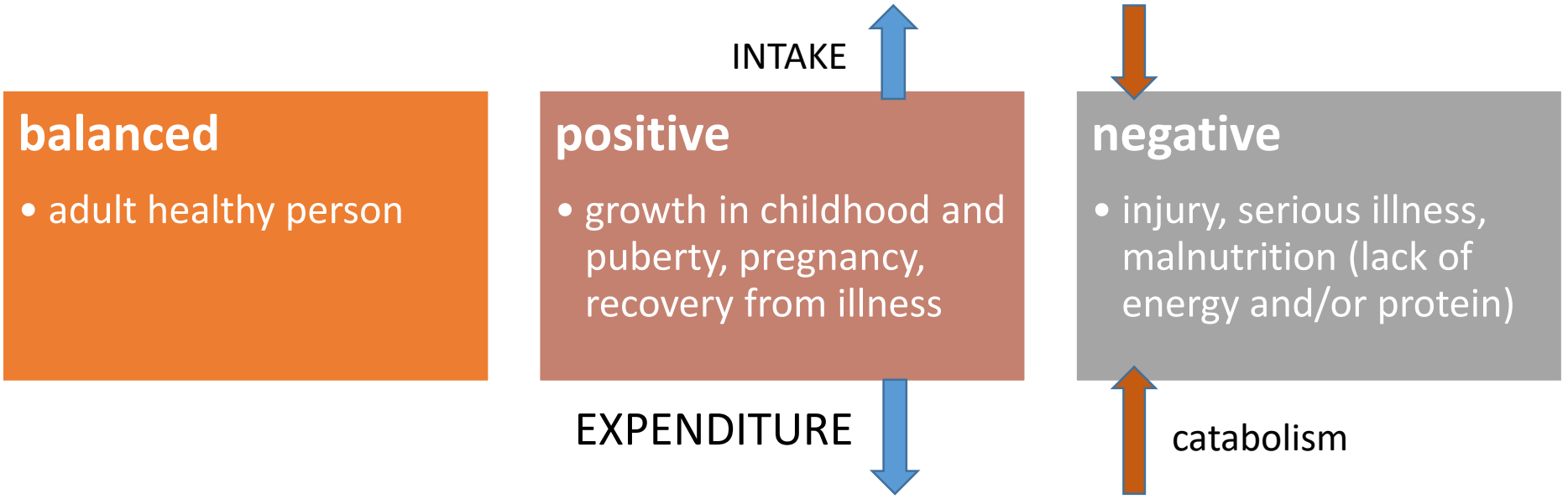


- NITROGEN BALANCE

Nitrogen balance

= nitrogen intake - nitrogen excretion

- nitrogen intake
  - food (enteral, parenteral nutrition)
- nitrogen excretion
  - URINE - urea, creatinine-> laboratory measurement
  - STOOL - exfoliated epithelia, digestive enzymes, undigested dietary proteins
  - SKIN - exfoliated epithelia



hunger center (VL) and satiety center (VM)

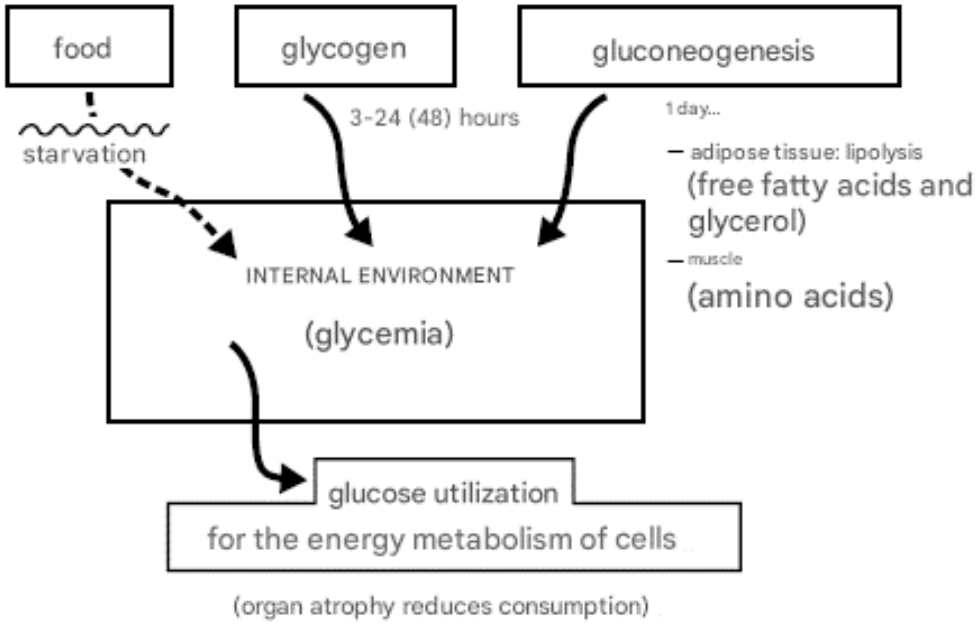
OREXIGENIC SUBSTANCES	ANOREXIGENIC SUBSTANCES
Ghrelin Peripheral organs, hypothalamus	**Cholecystokinin (CCK)** GIT
Neuropeptide Y(NPY) hypothalamus	**Glucagon-like peptide** GIT
Orexin A/orexin B Hypothalamus/ GIT	**Peptide YY(PYY3-36)** GIT
Endocannabinoid substances Hypothalamus, periphery	**Leptin** Adipose tissue, hypothalamus, brainstem
Somatoliberin	**Adiponectin,**
	Insulin

- complex interaction of neurotransmitters with either orexigenic (resulting in obesitogenic) or anorexigenic (leptogenic, i.e. slimming) effects
- The satiety center is stimulated immediately after a meal by stomach distension-> satiety center is also influenced by hormones released from the stomach wall, intestine and pancreas
- Suppression of appetite by the postprandial rise in blood glucose
- satiety regulation by protein intake
- sugars have only a short-term effect
- long-term regulation -the total body fat mass is used as a hormonally active tissue-> humoral signal sent by a fat cell is the hormone leptin. Leptin production is directly proportional to the amount of body fat.
- pain and some medications can decrease appetite

SIMPLE **starvation**

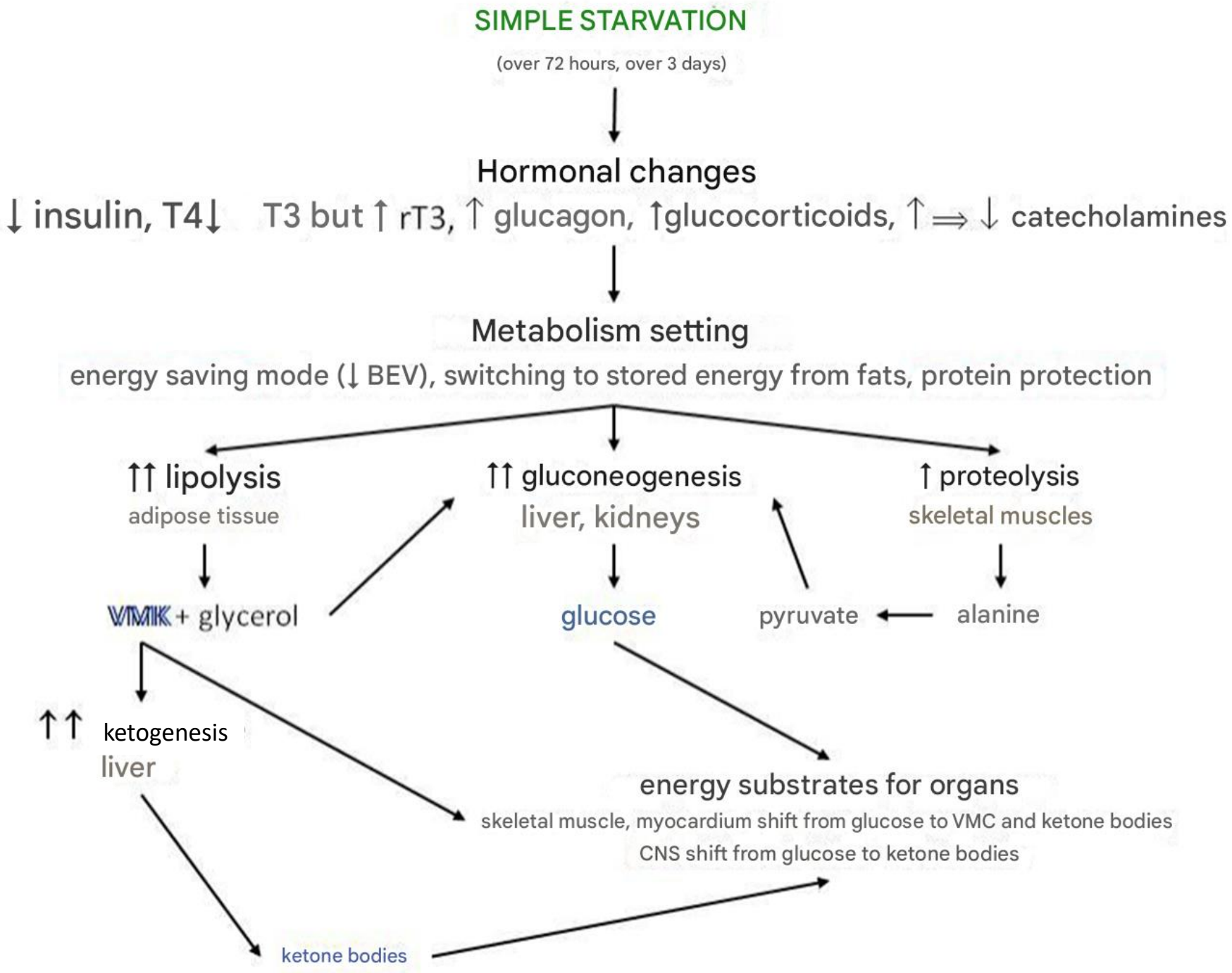
- **SHORT-TERM FASTING**
- Within 72 hours
- **glycogen** depletion (<48h)

- Catecholamines and GC
  - glycogenolysis, gluconeogenesis

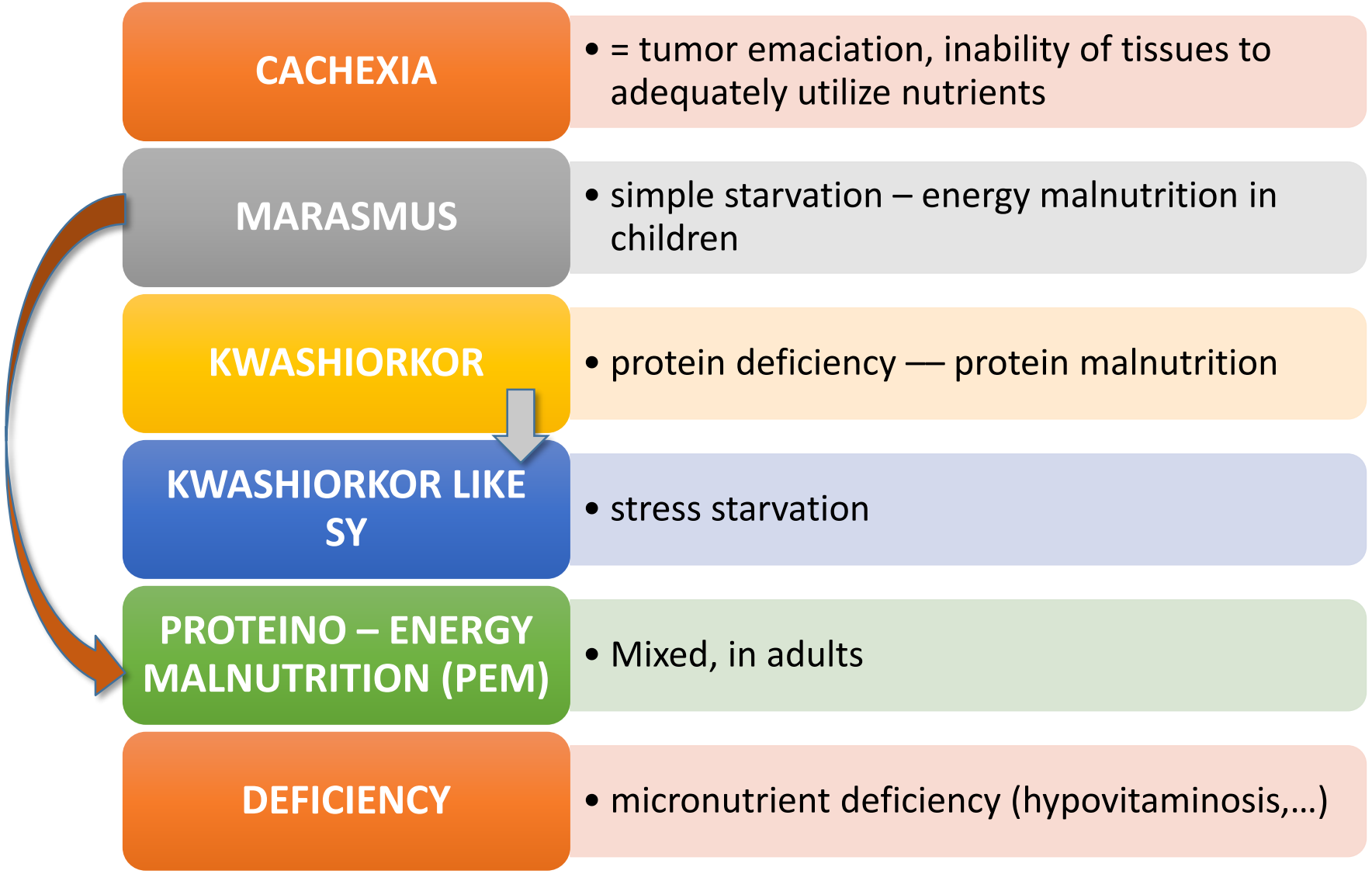


- **LONG-TERM FASTING**
- Adult patients can usually survive more than 3 weeks, consuming only water
- More than three days
- energy expenditure decreases by 40%
- DECREASE IN THYROID HORMONES
- DECREASE IN GC AND CA
- lipolysis increases
- ketone bodies in the liver (source for CNS) – decrease in pH
- proteins are saved

- 40th day proteocatabolism


SIMPLE STARVATION
(over 72 hours, over 3 days)
Hormonal changes
↓ insulin, T4↓ T3 but ↑ rT3, ↑ glucagon, ↑glucocorticoids, ↑ ⟹ ↓ catecholamines
Metabolism setting
energy saving mode (↓ BEV), switching to stored energy from fats, protein protection
↑↑ lipolysis
adipose tissue
↑↑ gluconeogenesis
liver, kidneys
↑ proteolysis
skeletal muscles
VMK + glycerol
glucose
pyruvate
alanine
↑↑ ketogenesis
liver
energy substrates for organs
skeletal muscle, myocardium shift from glucose to VMC and ketone bodies
CNS shift from glucose to ketone bodies
ketone bodies

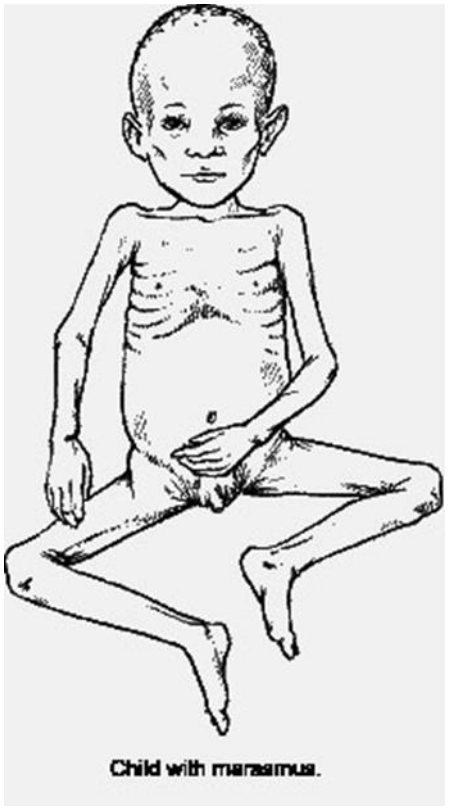
Malnutrition



MARASMUS

- Below 60% of weight for age
- malnutrition - balanced deficiency of essential nutrients and energy, i.e. proteins and fats
- uniform loss of subcutaneous fat and skeletal muscle
- simple starvation
- anorexia nervosa
- overall decreased metabolism
- TISSUE ATROPHY:
- BRADYCARDIA, HYPOTENSION
- dry thin skin, thin, falling hair
- MUCOSA ATROPHY:
  - Chronic diarrhea
  - SECONDARY MALABSORBATION – T and B immunodeficiencies, anemia, osteopathy
- patient dies from organ failure or infection

- NORMAL VISCERAL PROTEINS



Child with marasmus.

KWASHIORKOR

- diet poor in complete proteins-> deficiency of one or more essential amino acids

->DECREASE IN VISCERAL PROTEINS, LYMPHOCYTES

- Energy intake may be adequate-energy source is predominantly carbohydrates
- Normal appearance and weight, preserved fat stores
- children in developing countries (predominantly corn or rice diet lacks the amino acid tryptophan)
- Hypalbuminemia causes edema and ascites with general emaciation- Pitting edema with swelling of the ankles and feet
- Protruding belly- Enlarged liver with fatty infiltrates
- INFECTION, secondary immune disorders
- MUSCLE ATROPHY
- loss of appetite, vomiting, persistent diarrhea
- growth disorders, change in the quality of skin, hair, nails, teeth



kwashiorkor x marasmus

kwashiorkor

swelling of legs (oedema)

sparse hair

moon face, with little interest in surroundings

flaky appearance of skin

swollen abdomen

thin muscles, but fat present



marasmus

normal hair

old man or wizened appearance

thin limbs with little muscle or fat

very underweight body

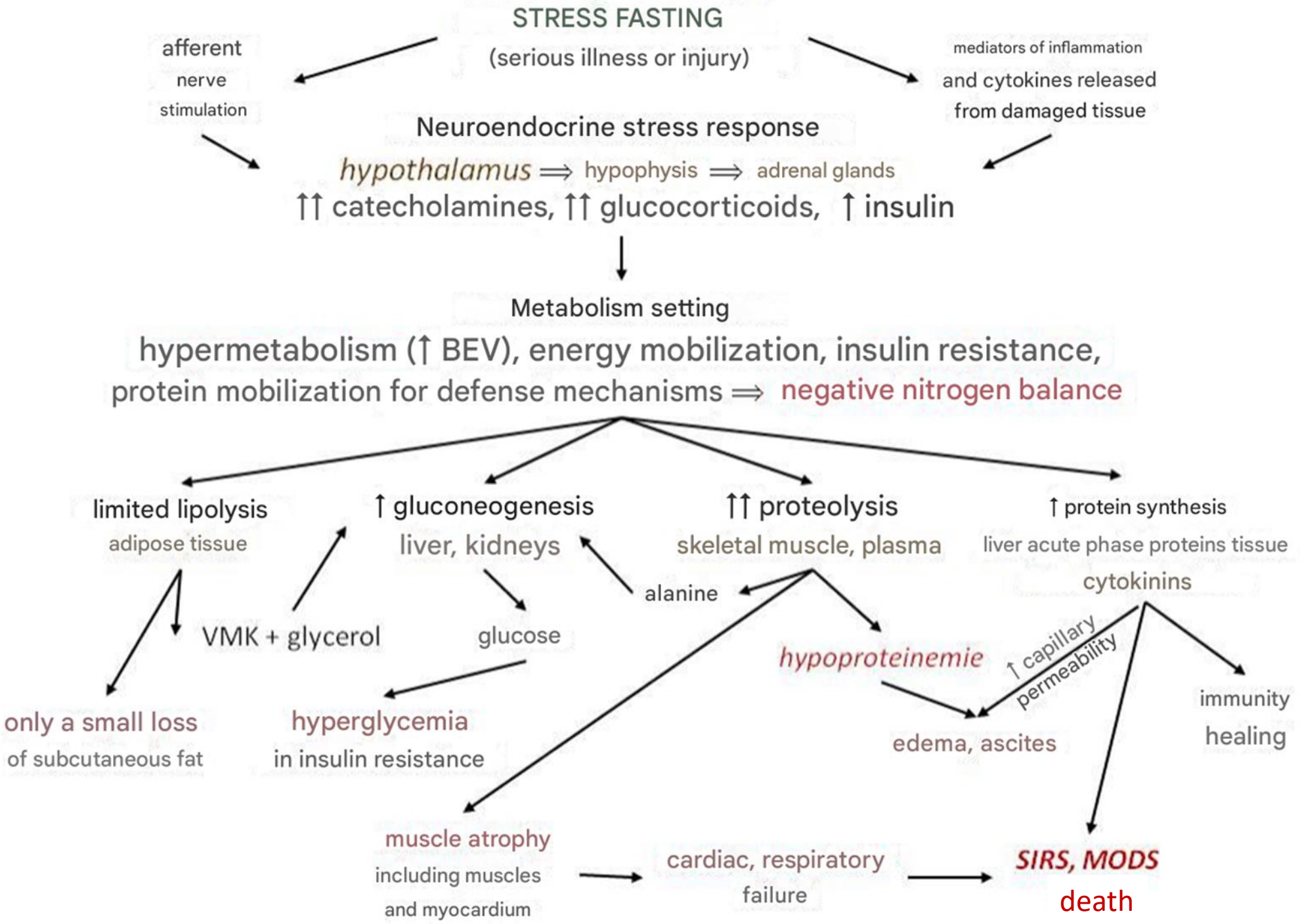


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kwashiorkor like sy – stress starvation

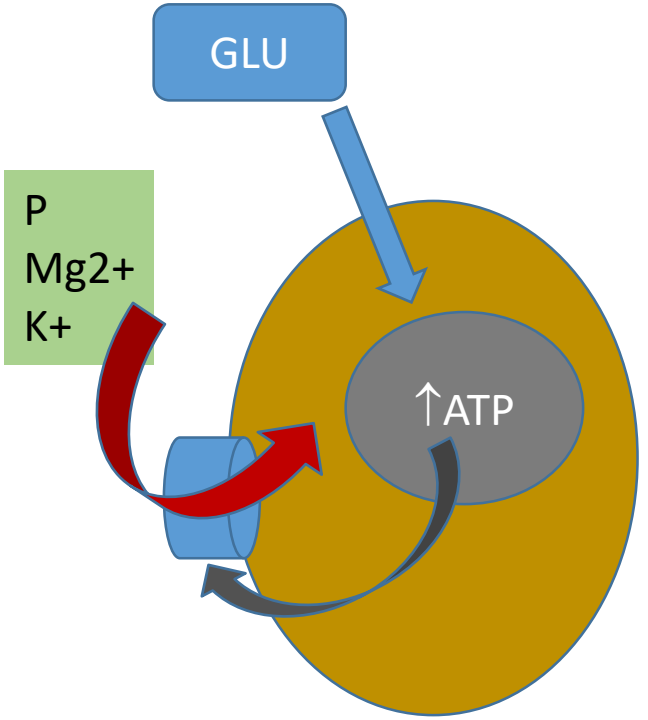
- develops rapidly
- Stress: GC
- IMMUNOSUPRESSION
- Intensive protein katabolism- LOSS OF MUSCLE MASS + VISCERAL PROTEINS
- HYPOALBUMINEMIA
- ECT expansion, fluid retention
- false weight gain
- hepatomegaly
- hyperglycemia, hyperinsulinemia - insulin resistance
- skin and hair changes

- Therapy: amk supply to reduce negative nitrogen balance


STRESS FASTING
(serious illness or injury)
afferent
nerve
stimulation
mediators of inflammation
and cytokines released
from damaged tissue
Neuroendocrine stress response
hypothalamus ⟹ hypophysis ⟹ adrenal glands
↑↑ catecholamines, ↑↑ glucocorticoids, ↑ insulin
Metabolism setting
hypermetabolism (↑ BEV), energy mobilization, insulin resistance,
protein mobilization for defense mechanisms ⟹ negative nitrogen balance
limited lipolysis
adipose tissue
↑ gluconeogenesis
liver, kidneys
↑↑ proteolysis
skeletal muscle, plasma
↑ protein synthesis
liver acute phase proteins tissue
cytokinins
VMK + glycerol
glucose
alanine
hypoproteinemie
↑ capillary
permeability
only a small loss
of subcutaneous fat
hyperglycemia
in insulin resistance
edema, ascites
immunity
healing
muscle atrophy
including muscles
and myocardium
cardiac, respiratory
failure
SIRS, MODS
death

Refeeding syndrom

- a set of metabolic abnormalities arising as a result of the restoration of food intake
- glucose + insulin into cells, restoration of ATP pump function



- decrease in serum levels of
- PHOSPHATES
- MAGNESIUM
- POTASSIUM

- *Manifestations:*
- muscle weakness
- water and sodium retention
- psychiatric changes and paresthesia
- up to malignant arrhythmia and heart failure

Anorexia nervosa

- psychogenic disorders of body image
- purposeful reduction in body weight by restricting food intake (sometimes associated with vomiting, abuse of diuretics, laxatives or thyroid hormones)
- chronic disease that leads to insufficient nutrient intake, secondary endocrine and metabolic changes and disruption of bodily functions.
- Physical manifestations correspond to malnutrition, but a striking difference is voluntary physical hyperactivity as an additional tool for maintaining low body weight.
- Body weight is reduced by more than 15% compared to normal, BMI>18.5
- All components of the sex hormone axis are suppressed
- Reduced estrogen production in women is manifested by amenorrhea and the long-term development of osteoporosis. Hypercortisolemia and insufficient calcium intake also contribute to osteoporosis.
- The suppression of the sexual axis in men leads to infertility and loss of libido. The production of thyroliberin (TRH), thyrotropin hormone (TSH), thyroxine (T4) and especially triiodothyronine (T3) is reduced, along with a corresponding increase in rT3 (the so-called low T3 syndrome). However, clinical manifestations typical of hypothyroidism are absent and the thyroid gland responds normally to a functional stimulation test.
- delayed gastric emptying, dilatation of the jejunum, and constipation.
- Higher death rate than bulimia



Orthorexia nervosa

- mental illness that can be translated as the right taste (from the Greek orthos = correct, orexis = taste). It is a pathological obsession with healthy nutrition and biologically pure foods, which is why it is also classified as an eating disorder.

Consequences

- the body is deprived of a large number of important substances and vitamins
- significant weight loss occurs
- they constantly think about food in their thoughts
- over time, they only consume food that they have prepared themselves
- they often convince those around them of the harmfulness of commonly available foods

bulimia nervosa

- Mental illness, WITHOUT WEIGHT LOSS
- episodes of overeating (bing eating), guilt
- VOMITING, LAXATIVES, DIURETICS ->LOSS OF K+
- Esophageal irritation – inflammation, erosion to bleeding
- Effect of NEUROPEPTIDES (cholecystokinin, endorphins, …)
- Teeth, nails tooth loss, brittle nails
- Internal environment
- METABOLICAL ALKALOSIS WITH HYPOKALEMIA, loss of Cl- and H+ arrhythmia
- Kidneys KALIOPENIC NEPHROPATHY
- GIT Degeneration of the myenteric plexus (laxatives), Mallory-Weiss Syndrome

malabsorption

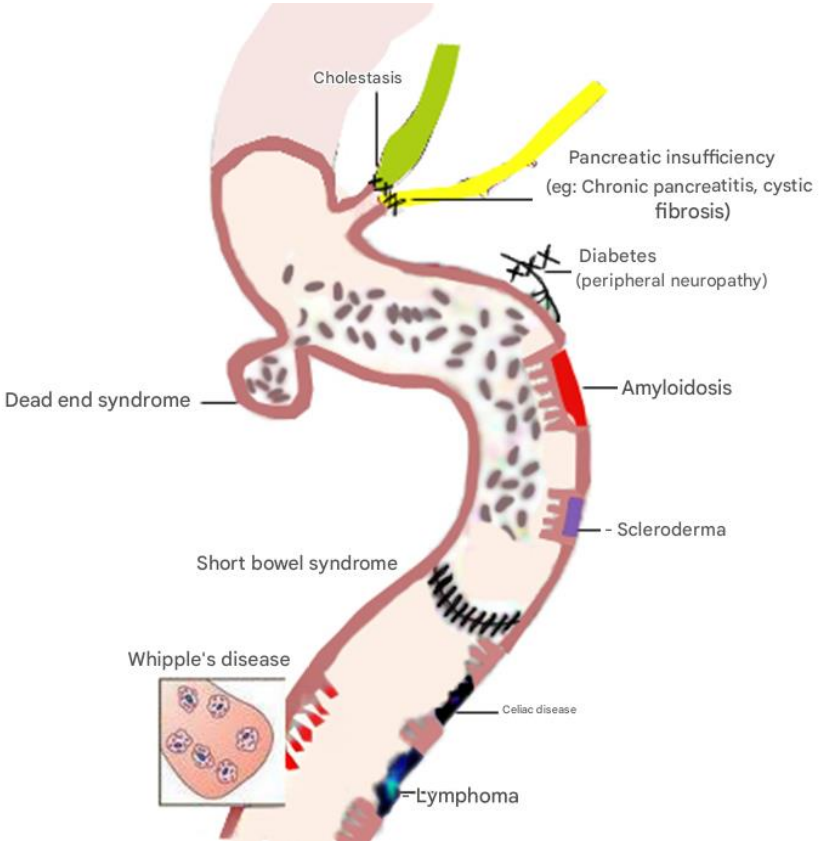
Celiac disease
Lactose intolerance

Malabsorption syndrome (malasimilatory)

- **MALDIGESCENCE**
- **Disorder of intraluminal DIGESTION**

- **MALABSORPTION**
- **Impaired ABSORPTION of nutrients**

- **Abnormal secretion of digestive juices**
- **Severe disturbance of motility of the digestive tract**
- **Insufficient food intake**
- **decreased resorption of both water and ions**
- **insufficient lipolysis**
- **Steatorrhea (impaired pancreatic secretion, accompanied by a lack of fat soluble vitamins)**



Malabsorption

- ***Primary***
- **at the enterocyte level**
- **COELIA DISEASE**
- **DISACCHARIDASE DEFICIENCY (lactase deficiency)**
- **TROPICAL SPRUE**
- **CARBOHYDRATE TRANSPORT DISORDERS (glucose and galactose malabsorption, fructose malabsorption)**
- **AMINO ACIDS TRANSPORT DISORDERS (cystinuria)**

- ***Secondary***
- **PANCREATIC MALDIGESTION AND MALABSORPTION** (disturbance in the pancreatic secretion)
- **WHIPPLE'S DISEASE**
- **blind loop syndrome**
- **SHORT BOWEL SYNDROME**
- **POSTRADIATION ENTERITIS AND COLITIS**
- **disturbance in the bile secretion**

Celiac disease, celiac sprue, gluten enteropathy

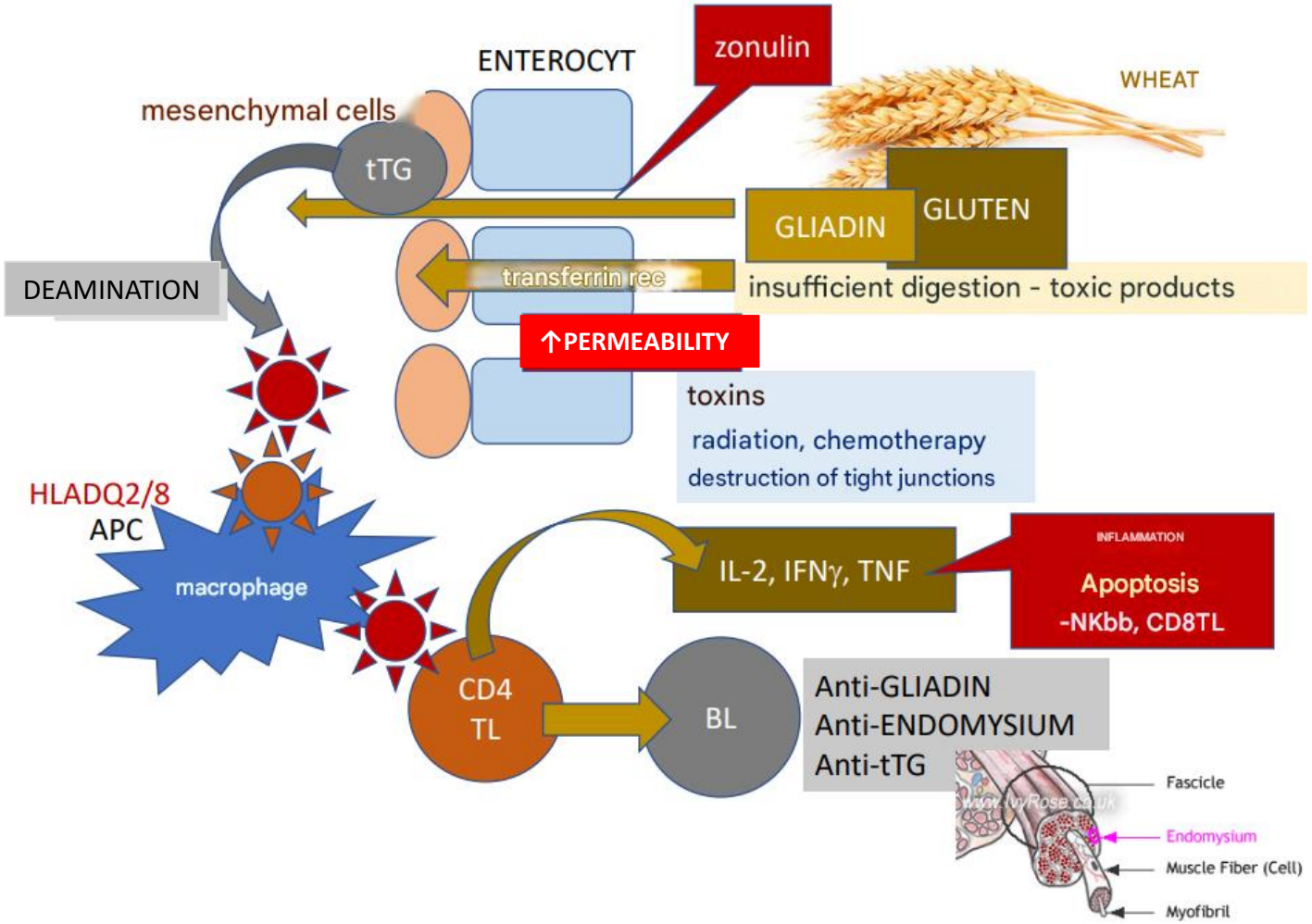
- ↑permeability of the intestinal mucosa to gliadin
- TISSUE TRANSGLUTAMINASE
- transfer of glutamyl residue, high specificity for gliadin

+ deamination: ->NEW PEPTIDES = antigen -> binding to HLA DQ2/8 (macrophage)

- T,B-LYMPHOCYTES
- proinflammatory cytokines, antibodies

➡ inflammatory reaction

- **DEVASTION OF JEJUNUM AND ILEUM**



DIARRHEA, FAILURE TO BRING, GROWTH ARRESTED

- In the 6th-18th month of life

Atypical:

- more than 50% may not have intestinal problems or diarrhea

? DERMATITIS HERPETIFORMIS DUHRING



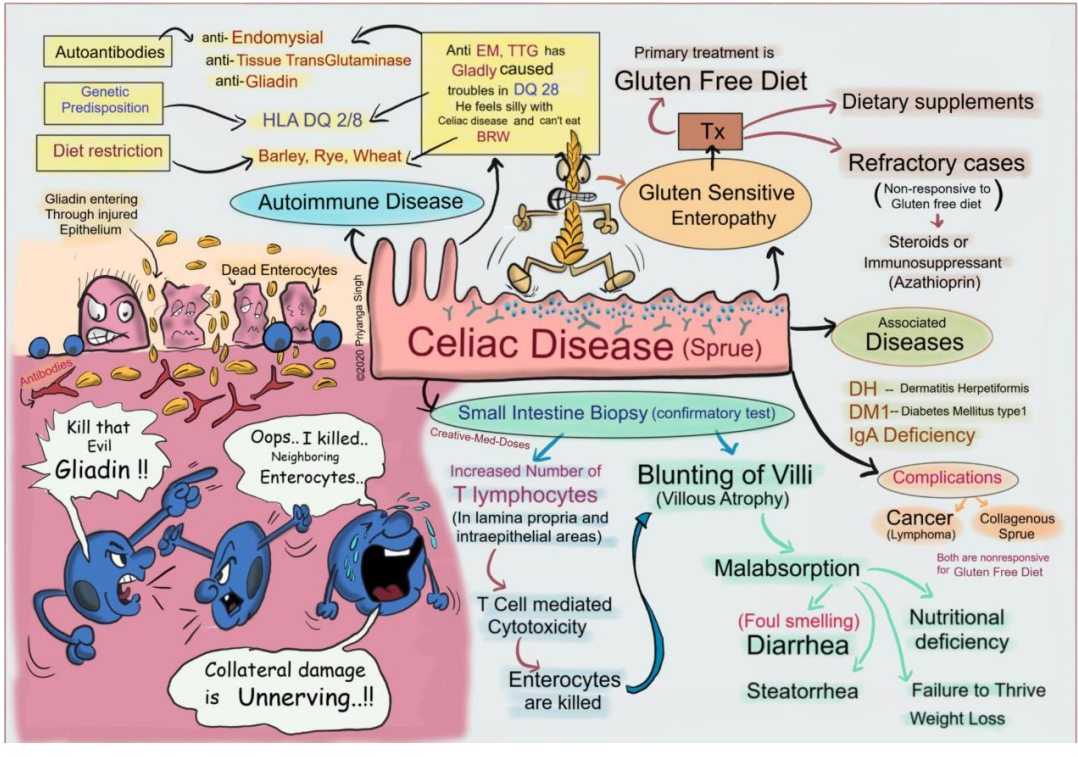


- bullous skin disease (IgA)
- iron deficiency anemia, growth retardation
- Hepatopathy, osteoporosis, intestinal findings consistent with celiac disease

ASYMPTOMATIC FORM

- no intestinal or extraintestinal symptoms
- lower physical performance,
- easier fatigue, tendency to depression,
- iron deficiency without anemia

- Bloating as a side effect of gluten
- GLUTEN-FREE DIET



LACTOSE INTOLERANCE

- LACTOSE = GALACTOSE + GLUCOSE
- selective malabsorption disorder
- Congenital deficiency X Acquired defect

❖ PRIMARY

gradual decrease in gene expression

❖ SECONDARY

decreased activity in intestinal diseases, ADMINISTRATION OF ATB

PLANT MILK

DISACCHARIDASE DEFECT – LACTASE

- inducible enzyme

activity increases after consumption of lactose

UNABSORBED LACTOSE

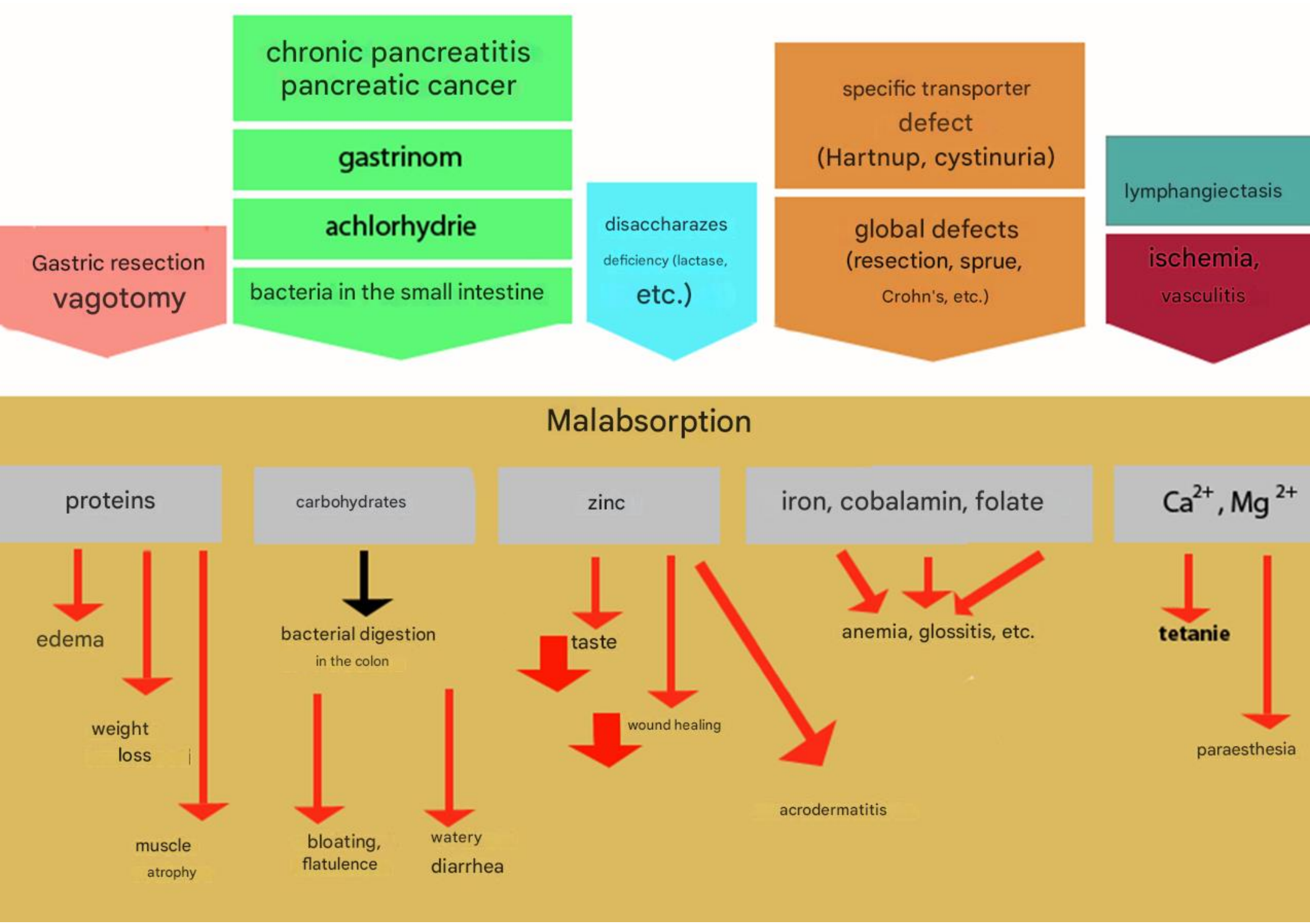
BACTERIAL MICROFLORA

LACTIC ACID, CO2 , CH4

- osmotic diarrhea, flatulence
- abdominal pain, sour-smelling stool

Clinically

- DIARRHEA
- WEIGHT LOSS
- WEAKNESS AND INEFFICIENCY
- ANEMIA
- EDUMS, MUSCLE ATROPHY
- HAEMORRHYMATIC DISORDERS
- SKIN, HAIR, NAIL DISORDERS
- MENSTURAL DISORDERS, INFERTILITY
- OSTEOPOROSIS, PARESTHESIA/TETANIA

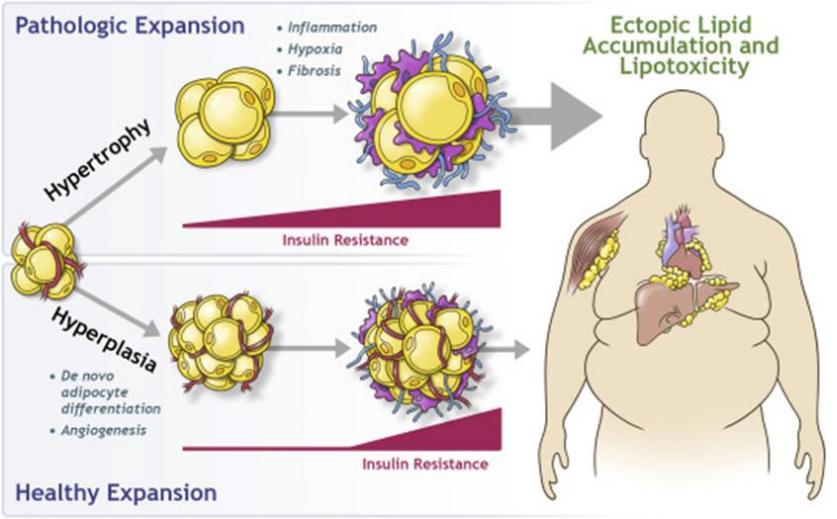


http://pfyziolklin.upol.cz/?p=4356

Obesity = hormonal imbalance

INSULIN
LEPTIN
ADIPONECTIN

GLUCOTOXICITY
LIPOTOXICITY



- Obesity is a syndrome characterized by excessive accumulation of energy reserves in the form of body fat.
- excess of fat reserves stored outside of physiological adipose tissue, which increases health risks and shortens life expectancy
- social, neuropsychic, metabolic and endocrine factors
- genetic predispositions 40%
- social and psychological (family tradition, preference for fatty and sweet foods, psychological stress)
- metabolic (various resting and postprandial energy expenditure, spontaneous physical activity, lipoprotein lipase activity, expression of β3-catecholamine receptors)
- endocrine (insulin production, insulin sensitivity, cortisol, growth hormone, sex hormones).

Primary-Risk periods -periods of hormonal changes –puberty, pregnancy and postpartum, menopause, andropause.

Secondary obesity

only 3–5%

- accompanies treatment with some drugs that interfere with lipid and energy metabolism and central regulation of food intake
- hormones with lipo-anabolic effect consists of cortisol, insulin and estrogens. Their pathological excess is therefore often associated with obesity. Lipo-catabolic hormones include glucagon, STH and catecholamines (acting via β2 and β3 receptors).
- Cushing's syndrome – hypercortisolism manifests itself both by increasing appetite and increasing visceral fat deposition-moon-shaped face, fat deposition on the neck, supraclavicular and abdominally.
- Hypothyroidism – obesity is associated with reduced energy consumption; however, hypothyroidism does not lead to severe obesity, the weight gain does not exceed 5–10 kg.
- Insulinoma – hyperinsulinemia stimulates food intake with recurrent hypoglycemia (however, most patients with islet tumors of Langerhans are not obese).
- Hypogonadism.
- Hypothalamic obesity is a rare type of obesity in which the function of the ventromedial centers of the hypothalamus is genetically determined or acquired (tumors, radiation, surgery, inflammation) is altered.
- Obesity develops in childhood and is the result of a combination of dysfunction of the hypothalamic centers of food intake and hormonal disorders promoting obesity (hypogonadism, STH deficiency)
- There are also monogenic types of obesity resulting from a genetically determined disorder of the formation of a mediator or its receptor, leptin, leptin receptor, POMC, etc.

obesity in children

- Predisposing factors and lifestyle
- Intrauterine development and
- prepuberty and puberty, especially in girls.

- early onset of its metabolic complications.
  - Hyperlipidemia
  - arterial hypertension
  - Hepatic steatosis
  - impaired glucose metabolism

Endocrine activity of adipose tissue and systemic inflammation

- fat cells -> the target organ of cortisol, insulin, STH, sex hormones, thyroid hormones, etc.
- In men and postmenopausal women, is adipose tissue the main source of estrogens, which block osteoclastic bone resorption and slow down the development of osteoporosis
- In obese women, excess estrogen can lead to ovarian cycle disorders.
- Until menopause, progesterone largely eliminates the carcinogenic effect of estrogen on estrogen-dependent organs, especially the uterus -> obese women: persistent estrogen production in adipose tissue increases the risk of breast and uterine cancer
- The production of adipokines with insulin-resistant effects and pro-inflammatory cytokines increases when the physiological capacity of adipose tissue to store lipids is exceeded.
- Hypertrophied adipocytes die to an increased extent and provoke an inflammatory response including increased accumulation of inflammatory cells in adipose tissue.
- Pro-inflammatory cytokines released from adipose tissue cause low-intensity systemic inflammation, but with long-term effects on the body, including the acceleration of atherosclerosis.

metabolic syndrome

- obesity (abdominal);
- insulin resistance (type 2 diabetes mellitus);
- arterial hypertension;
- hyperlipoproteinemia;
- hypercoagulable state
- The central part of the metabolic syndrome is insulin resistance
- Visceral fat is more metabolically active
- Insufficient glucose uptake by muscle cells due to a reduced amount of GLUT4 on their membrane and increased gluconeogenesis in the liver cause hyperglycemia.
- Hyperglycemia increases insulin secretion-> hyperinsulinemia
- Insulin inhibits lipolysis in adipose tissue and insulin resistance is thus also linked to android-type obesity and its metabolic and cardiovascular complications.
- When the beta cells of the islets of Langerhans are unable to compensate for insulin resistance with sufficient hyperinsulinemia, type 2 diabetes mellitus develops.
- Hyperinsulinemia predisposes to arterial hypertension by increasing sodium retention in the kidneys, stimulating the formation of vasoconstricting endothelin and, conversely, reducing the production of (vasodilating) nitric oxide in the endothelium, and by increasing the tone of the sympathetic nervous system.
- The main complications of metabolic syndrome therefore include cardiovascular disease due to accelerated atherosclerosis and a procoagulant state, as well as type 2 diabetes mellitus and some malignant tumors due to hyperinsulinemia.

Disorders of purine, pyrimidine and sugar metabolism

Purine metabolism

– adenine, guanine

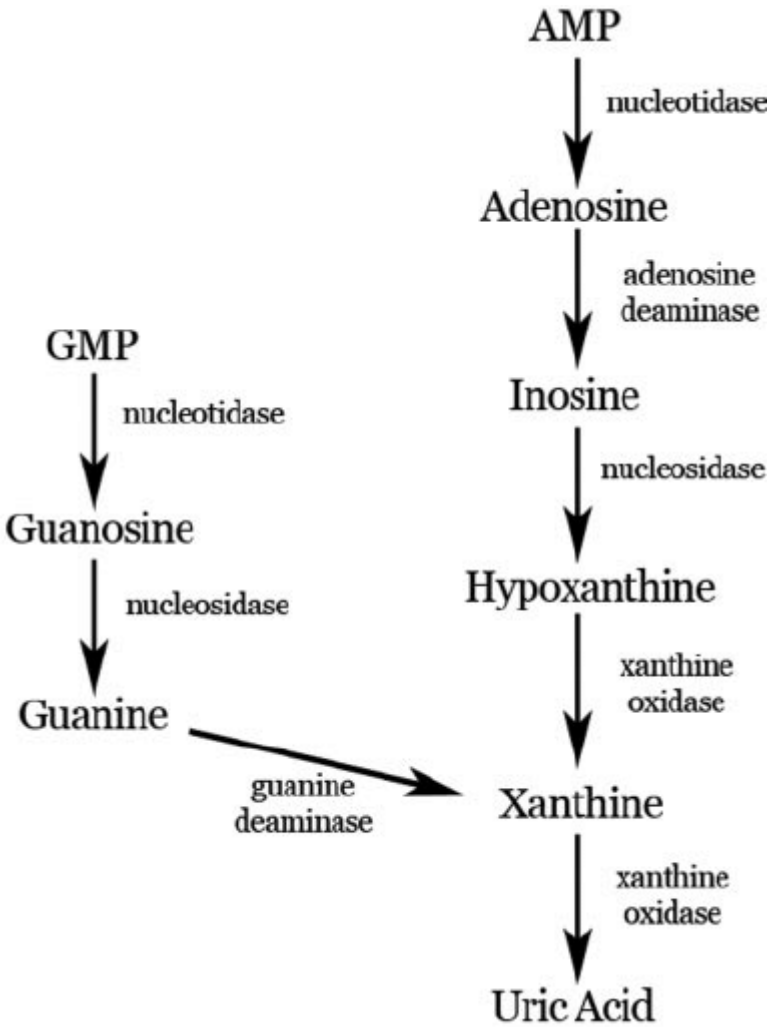
Xanthine oxidase (XO)

adenine → hypoxanthine → xanthine

guanine → xanthine → uric acid

part of hypoxanthine and xanthine is converted back to nucleotides and utilized

Hypoxanthine-guanine-phosphoribosyltransferase HGPRT



Hyperuricemia

= increased concentration of uric acid in the blood

- from food
- purine metabolism

congenital

- ENZYMATIC DEFECT HGPRT (Lesch-Nyhan syndrome – X-linked on.)
  - Hypoxanthine-guanine phosphoribosyltransferase
  - ↑phosphoribosyl diphosphate PRDP: does not enter nucleotide synthesis
  - Encephalopathy, spastic paralysis, self-harm

acquired

- ↑ NUCLEOTIDE CATABOLISM
  - cytostatics, more severe catabolic states, fructose
- ↑ DIETARY SUPPLY
- ↓URIC ACID EXCLUDING
  - renal insufficiency, drugs – thiazide diuretics, alcohol
  - influence of urine pH – better excreted in alkaline X lactate

GOUT

- more common in men and in people with a high meat intake
- combination of various factors (increased synthesis, selective transport disorders at the level of the renal tubules)
- heterogeneous group of disease states whose common feature is hyperuricemia and formation + deposition of urate crystals in tissues, especially in joints and their surroundings primary – idiopathic x secondary
- Microcrystals are phagocytosed by leukocytes and thus trigger a cascade of local inflammatory processes

- Most affected:
- JOINTS
- KIDNEYS
  - urate nephrolithiasis, tubulointerstitial nephritis sometimes leading to renal insufficiency
- CIRCULATORY SYSTEM
  - acceleration of atherosclerosis, more frequent thrombosis
- eyes, skin, more frequent DM2

Acute gout attack

Crystals falling out into the joint cavity (reduced temperature and reduced pH at night)

subsequently coated with IgG

INFLAMMATION

phagocytosis neutrophils + neutrophil damage crystals

release of enzymes → TISSUE DAMAGE



90% monoarticular -> 70% 1st MTP joint

- sudden onset (often at night), severe pain,
joint red, hot, hypersensitive,
general symptoms
- average duration of attack 1 – 2 weeks

Clinical stages of gout

1. Asymptomatic hyperuricemia	• ↑ S-KM • 90% men, 5th-6th decade • postmenopausal women (uricosuric effect of estrogens)
2. Acute gout attack	• provoking moment: dietary excess, stress • ACUTE GOUTY ARTHRITIS
3. Intercritical period	• rest intervals between attacks ½ - 2 years
4. Chronic tophic gout	• TOPHOSIC DEPOSIT: cartilage, synovium, tendons, skin • urate nephropathy

Secondary hyperuricemia

- Increased turnover or breakdown of nuclear elements
  - Myelo- and lymphoproliferative diseases, increased production of erythroblasts (hemolytic and macrocytic anemias), cytostatics, crush syndrome
- Drugs
  - Cytostatics, corticosteroids in high doses (catabolism), blocking urate secretion – diuretics (furosemide)
- Renal insufficiency
  - Reduced glomerular filtration
- Chronic alcoholism
  - Increased purine intake, ↓pH, lactate reduces excretion by the kidneys
- Chronic lead poisoning

Urate nephropathy

- **1. URATE UROLITHIASIS**
- formation and presence of urate stones in the urinary tract
  - 10-25% of patients with gout
  - risk of precipitation: ↓urine pH, dehydration

→ RENAL COLICS

- **2. INTERSTITIAL GOUTY NEPHRITIS**
- deposition of Na-urate crystals in the renal interstitium with subsequent inflammatory changes

→ albuminuria, ↓kidney concentration capacity, renal hypertension

→ CHRONIC RENAL INSUFFICIENCY

- **3. ACUTE URATE TUBULOPATHY**
- during radio or chemotherapy hematomalignancy
- ->**URINARY TRACT OBSTRUCTION**

→ acute tubular necrosis (distal tubules and collecting ducts)

→ ACUTE RENAL FAILURE

Disorders of pyrimidine metabolism

Pyrimidine synthesis disorders

- UMP synthase deficiency (Orotic aciduria)
- AR heredity
- Orate accumulation
  - **Crystalluria**

Pyrimidine deficiency

- **megaloblastic anemia**
- **growth disorders, psychomotor retardation, leukopenia, malaise**

Pyrimidine degradation disorders

- Dihydropyrimidine dehydrogenase (DPD) deficiency
- AR inheritance
- Uracil and thymine accumulation

Complete deficiency in children

- **epilepsy, mental retardation, autism, microcephaly**

Partial deficiency

- **manifestations of 5- flurouracil therapy (tumors)**
- **neutropenia, stomatitis, neurological symptoms**

Glycogenosis

- Storage polysaccharide
stored in the cytoplasm of
healthy cells.

- Stored in the liver and
muscles.
- Can only be released into the circulation from the liver
(glucose-6-phosphatase).

- Glycogenoses
- = genetically determined
disorders of
glycogen metabolism
Glycogen accumulates in the cells in the form of
granules.

https://www.youtube.com/watch?v=LuVcPNF5Slg

Glycogenosis

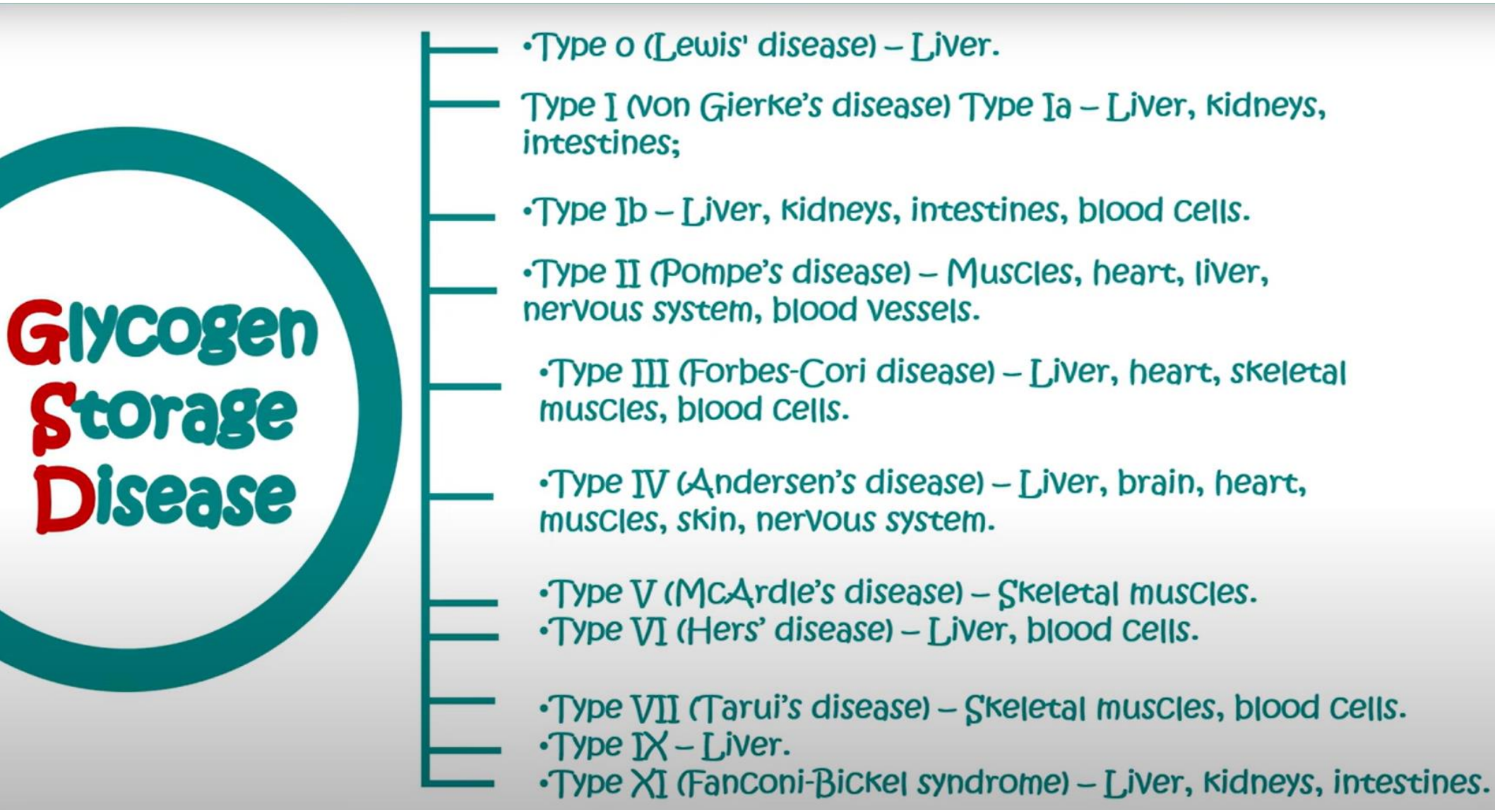
- Hepatic: GSD I, III, IV, VI
- Hepatomegaly
- Hypoglycemia (liver = main source of glucose! has glucose-6-phosphatase)

↑ glucose-6-phosphate = ↑ lactate

lipolysis → hyperlipidemia and ketogenesis → KAC

- Myopathic: GSD V, VII
- Glycogen accumulation in muscle
- Muscle weakness
- Muscle cramps
- Necrosis (myoglobinuria)

Other: GSD II
Involvement of other organs



https://www.youtube.com/watch?v=yvaqtsSceww

Von Gierke's disease (GSD I)

- deficiency of the enzyme glucose-6-phosphatase → hypoglycemia (→ cramps) and lactic acidosis
    - AR, incidence of GSD I in the general population is 1/100,000, Ashkenazi Jewish population has 5 times higher prevalence
    - G6Pase enzyme is primarily expressed in liver, kidneys and intestines →accumulation of glycogen and fat in liver, kidneys and intestinal mucosa
- hepatomegaly
- nephromegaly
- ↑ uric acid
- ↑ lipids

Pompe disease (GSD II)

↓ Lysosomal α1,4 glucosidase

- AR inheritance
- 1:40,000
- Deficiency of lysosomal enzymes + glycogen storage in skeletal and cardiac muscle

→ MUSCLE HYPOTONY (at any time during life)

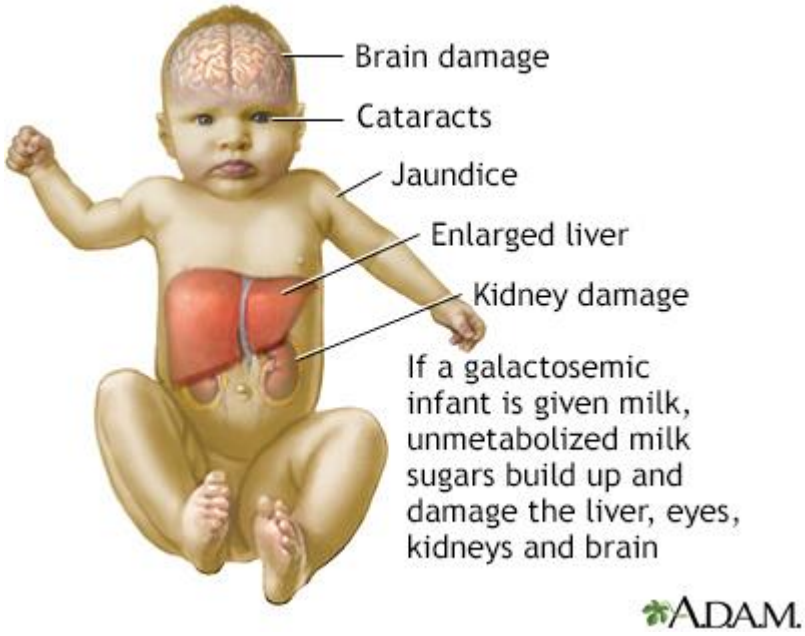
❖ Infantile type

skeletal muscle → hypotonia, failure to thrive, respiratory muscle weakness → sleep apnea; myocardium → hypertrophic cardiomyopathy; vascular endothelium, astrocytes, liver → hepatomegaly, kidneys

❖ Late type (juvenile and adult)

muscle weakness of the pelvic ring and proximal muscle groups of the limbs, exertional dyspnea, apnoeic pauses, feeding disorders (swallowing)

Classical galactosemia



https://www.mountsinai.org/health-library/diseases-conditions/galactosemia

- AR heredity
- incidence is 1:35,000–1:50,000
- galactose-1-phosphate uridyltransferase deficiency

between 4th–9th day after milk intake:

❖ vomiting, diarrhea, milk refusal
❖ hepatic insufficiency with jaundice and hepatomegaly
❖ hemorrhagic diathesis
❖ lethargy, convulsions
❖ pseudotumor cerebri, brain edema
❖ bilateral cataract (↑ galacticoll)

- PICTURE OF SEPSIS WITH LIVER AND RENAL FAILURE
- speech disorder
- in girls hypergonadotropic hypogonadism

Evidence of increased galacticoll concentration in urine

Disorders of protein and amino acid metabolism

Protein metabolism

digestionprotein

catabolism

proteins → amino acids → carbon skeleton

amino acid metabolism

removal of the amino group

pyruvate
acetyl CoA



gluconeogenesis
lipogenesis

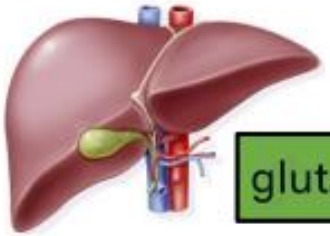
new substances

citrate cycle intermediates

citrate cycle + respiratory chain

H_2O
CO_2
ATP







glutamine



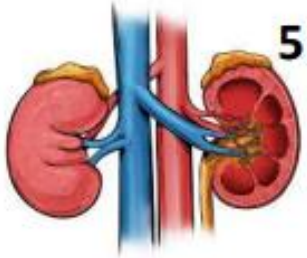
+ glutamate

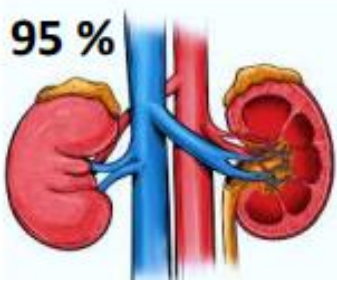
5 % NH_3/NH_4^+

ammonia

ureosynthetic cycle

urea





95 %

Phenylketonuria (PKU)

- 377 / 5 000
- AR, in the Czech Republic 1:6500, world 1:13000
- absence or reduced activity (600 gene mutations on chromosome 12 described) of phenylalanine hydroxylase = hydroxylation of Phe to Tyr.
- alternative degradation of Phe – phenylpyruvate (transamination), phenyllactate, phenylacetate or phenylethylamine is formed
- -> mouse-like urine odor
- Tyr deficiency -> mental retardation, epilepsy, reduced pigmentation
- newborn screening



PHENYLALANINE HYDROXYLASE

phenylalanine → tyrosine → dopamine

tyrosine → catecholamine

tyrosine → melanin

tyrosine → thyroxine

tetrahydrobiopterin BH4 ⇄ dihydrobiopterin BH

GTP

dihydropteridine reductase

phenylalanine → phenylpyruvate

phenylpyruvate → phenyl lactate

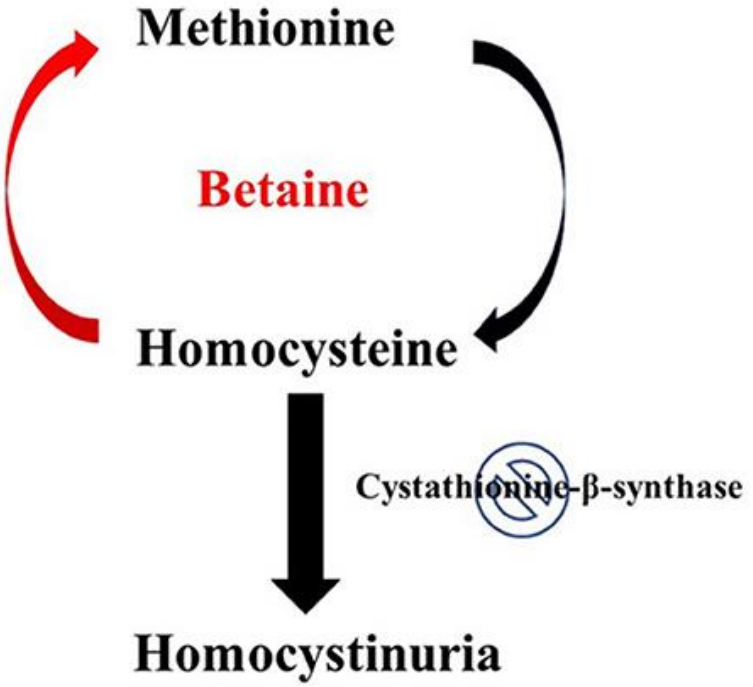
phenylpyruvate → phenylacetate





Homocystinuria

- AR
- a metabolic disease that is caused most often by a disorder of the activity of cystathione beta-synthase (CBS deficiency), an enzyme involved in the body in the conversion of the amino acid methionine to cysteine.
- In addition to the amino acid METHIONINE, the very toxic HOMOCYSTEINE also accumulates
- 1:83,000 to 1:15,000

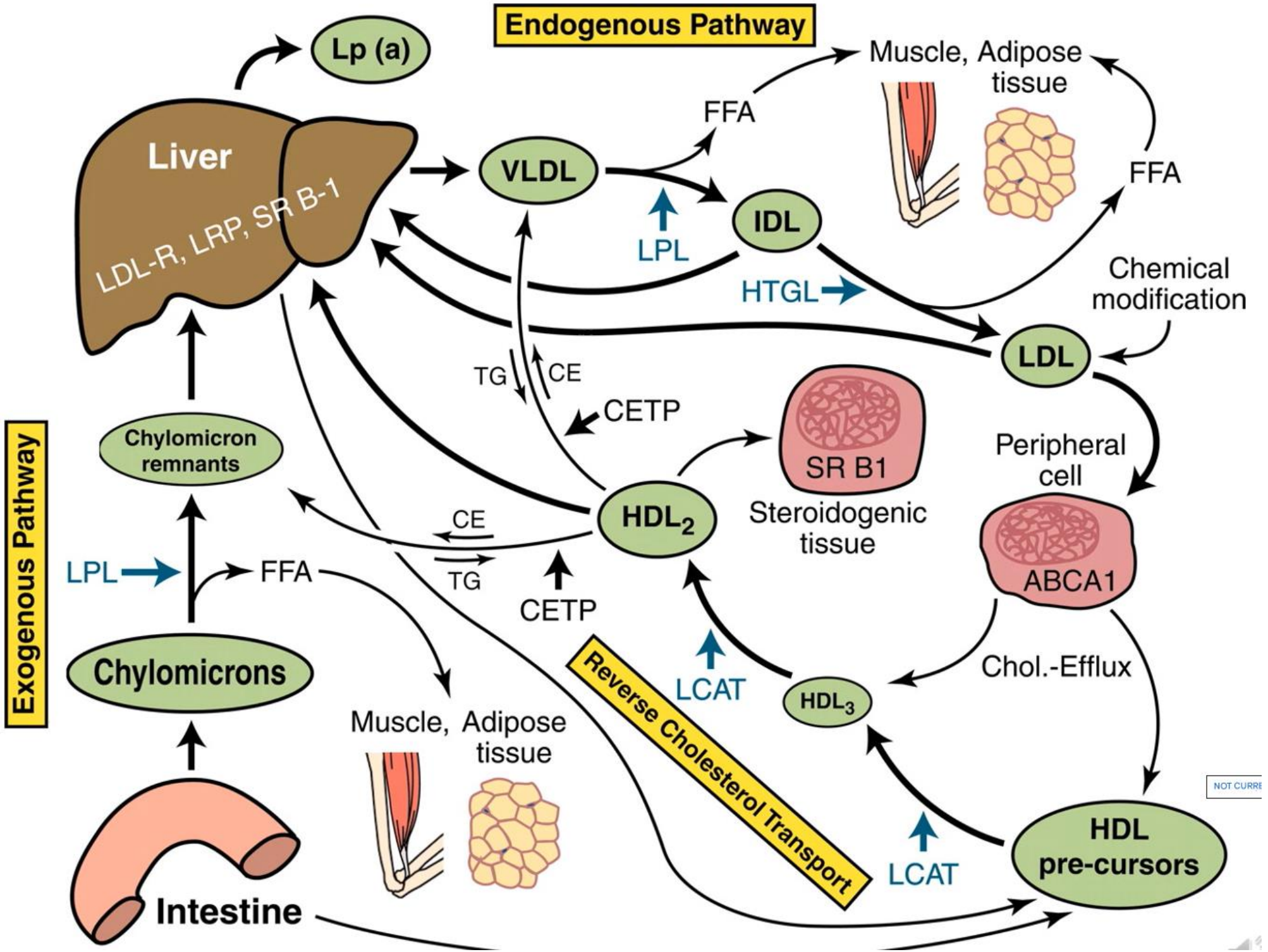


https://www.frontiersin.org/journals/nutrition/articles/10.3389/fnut.2021.690359/full

Homocystinuria

- Eye disorders:
  - lens dislocation, myopia, glaucoma, retinal detachment, optic nerve atrophy, cataracts
- Bone disorders:
  - metabolic osteopathy, osteoporosis, scoliosis, pathological fractures and vertebral collapse
- Central nervous system disorders:
  - developmental delay and mental retardation (60%), seizures (50%), EEG abnormalities and psychiatric disorders
- Vascular disorders:
  - thromboembolic complications (deep vein thrombosis, pulmonary embolism, stroke)

DISORDERS OF LIPID AND LIPOPROTEIN METABOLISM


Endogenous Pathway
Lp (a)
Liver
LDL-R, LRP, SR B-1
VLDL
FFA
Muscle, Adipose tissue
FFA
LPL
IDL
HTGL
Chemical modification
LDL
TG
CE
CETP
SR B1
Steroidogenic tissue
Peripheral cell
ABCA1
Exogenous Pathway
Chylomicron remnants
HDL2
CE
TG
CETP
LPL
FFA
Chylomicrons
LCAT
HDL3
Chol.-Efflux
Reverse Cholesterol Transport
Muscle, Adipose tissue
LCAT
HDL pre-cursors
Intestine

Dyslipidemia (or dyslipoproteinemia)

- Abnormal concentrations of serum lipoproteins

  Causes:

  - Primary dyslipoproteinemias result from:
  - genetic defects (familial dyslipoproteinemias)
  - dietary factors

- Secondary dyslipoproteinemias result from diabetes, hypothyroidism, pancreatitis, renal nephrosis and medications such as certain diuretics, beta-blockers or glucocorticoids.

Lipoproteins

- Chylomicrons (CM) -formed in the intestine and transport absorbed exogenous triacylglycerols. The main apoprotein is B48.
- VLDL (Very Low Density LP, pre-ß-LP) - transport endogenous triacylglycerols synthesized in the liver.
- IDL (Intermediate-Density LP) are formed from VLDL by the action of lipoprotein lipase and have a lower content of triacylglycerols with a relatively higher proportion of cholesterol esters.
- LDL (Low Density LP, ß-LP) contain mainly cholesterol esters and a single apoprotein (B 100).
- HDL (High Density LP, α-LP) reverse cholesterol transport. The main apoprotein is apoprotein A. HDL cholesterol (HDL-C) is an important protective factor in atherosclerosis; its low concentrations are one of the components of the metabolic syndrome.
- Lipoprotein(a), Lp(a) is a lipoprotein particle of LDL with a molecule very similar to plasminogen attached. It is a significant risk factor for atherosclerosis. The liver is the key organ for LP metabolism. Lipoproteins are produced in it (“intrinsic pathways”) and it is the only organ capable of eliminating cholesterol from the body, either directly into bile or after its conversion to bile acids. These are also part of bile, but undergo enterohepatic circulation. In addition to the liver, lipoproteins are also produced in the intestine (“extrinsicpathways”).
- Molecules important for LP metabolism are apolipoproteins, enzymes and cell receptors.

Dyslipidemia

- increase in the concentration of some lipoproteins in plasma
- pathogenesis of atherosclerosis and its complications.

The cause is disorders of:

- enzymes
- receptors
- apolipoproteins
- excessive synthesis of own lipids

- Dyslipidemias are genetically determined-> complex genetic influences combine with external factors
- hyperlipidemia -increased concentration of lipids (triacylglycerols and cholesterol) in plasma.
- Hypercholesterolemia -increased concentration of cholesterol in the blood.
  - Blood cholesterol is made up of up to 75% cholesterol contained in LDL -atherogenic.
- HDL also contain cholesterol, but from the point of view of the development of atherosclerosis, it acts as a protector with its function of reverse cholesterol transport.
- Hypertriacylglycerolemia -increase in the concentration of triacylglycerols in the blood.
  - risk factor for pancreatitis and an independent factor for the development of atherosclerosis

Primary dyslipidemia

- monogenic or polygenic inheritance
- referred to as "familial".
- Familial combined hyperlipidemia - elevated cholesterol and triglycerides.
- Familial hypercholesterolemia LDL receptor mutation. .
- Familial apo B100 defect Familial hypertriacylglycerolemia

Secondary dyslipidemia

- diabetes mellitus;
- nephrotic syndrome and chronic renal failure;
- hypothyroidism;
- primary biliary cirrhosis;
- alcoholism (not to be confused with the effect of small doses of alcohol);
- the effect of medications (e.g. oral contraceptives).

Hypolipoproteinemia

Familial hypo-β-lipoproteinemia

- rare genetic abnormality, probably with autosomal dominant inheritance.
- ↓LDL cholesterol in plasma
- longevity, probably due to the low incidence of myocardial infarction

Abetalipoproteinemia

- rare autosomal recessive disease in which neither Apo-B48 is synthesized in enterocytes nor apo-B100 in the liver.
- chylomicrons are absent in the circulation and the transport of endogenous cholesterol to peripheral cells via LDL is impaired, concentration of cholesterol in plasma is reduced.
- Impaired absorption of lipids from the intestine causes their excessive loss in the feces and thus a deficiency of fat-soluble vitamins.
- delayed development and reduced intelligence, polyneuropathy, and muscle weakness.

Hypoalphalipoproteinemia

- Conditions with reduced HDL-cholesterol have an increased risk of atherosclerosis and the resulting cardiovascular disease. The familial form appears to have dominant inheritance.
- The result is a relative lack of apoC-II and increased VLDL levels.

Analphaloproteinemia (Tangier disease)

- It is a rare disease with autosomal recessive inheritance, characterized by a complete absence of HDL in plasma.

Recommended resources

- Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014. Chapter 10.
- McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018. Chapter 16.
- Silbernagl S, Lang F. Color Atlas of Pathophysiology, 3rd Ed. Thieme, 2016.
- Nečas a spolupracovníci, *Obecná patologická fyziologie*. Karolinum, 2000